



BITS Pilani
Pilani Campus

Machine Learning

AIML CZG565

Unsupervised Learning

Course Faculty of M.Tech Cluster
BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Profs.Sugata, Chetana, Rajavadhana, Monali, Seetha, Vimal, Swarna

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction & Mathematical Preliminaries
M2	Machine Learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

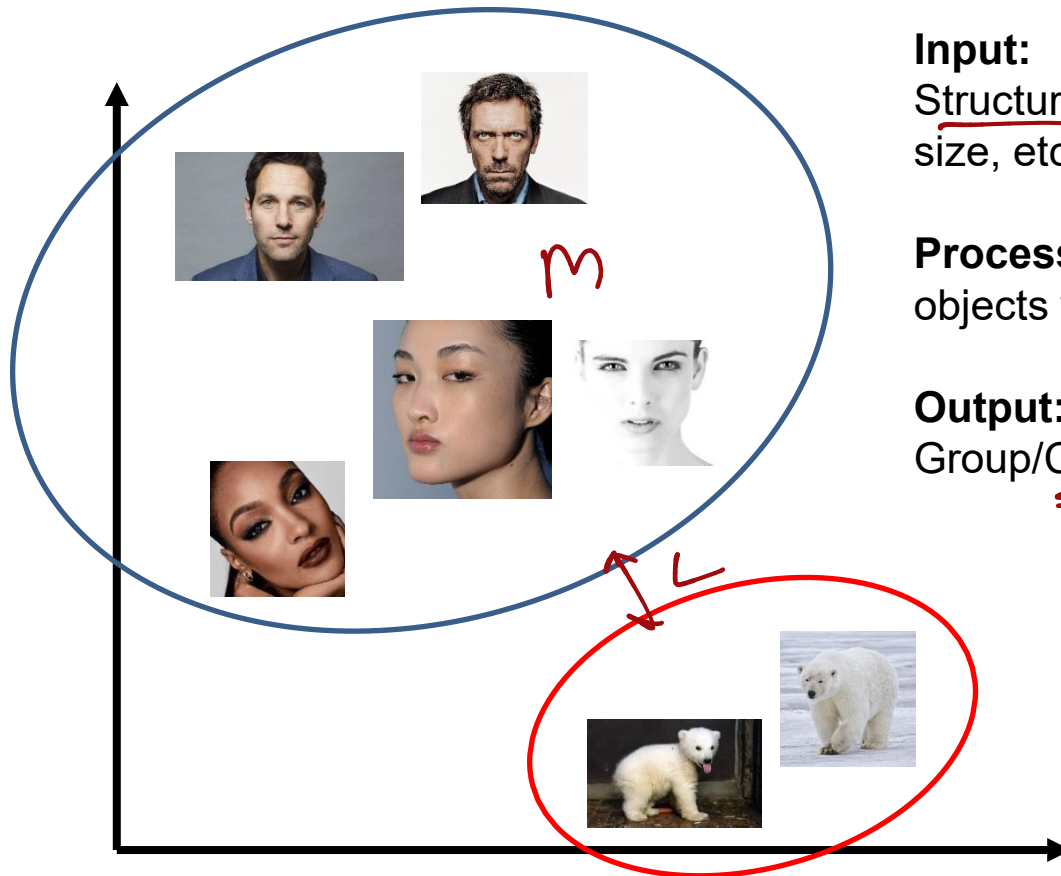
Module –Unsupervised Learning

- K-means Clustering ✓
- Mixture Models ✓
- Expectation Maximization (EM) Algorithm

Algo. ← Unlabeled data
↓
WE ← categorize

Unsupervised discovery

Task : Group the objects based on the characteristics features



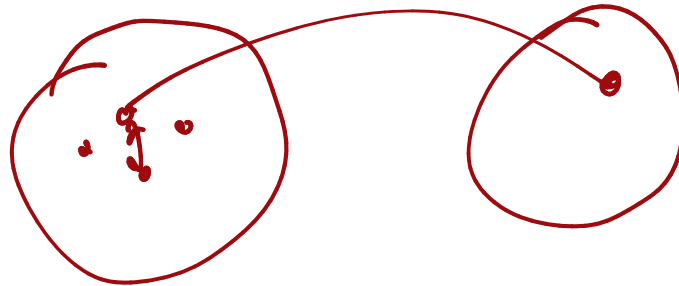
Input:

Structured data of features like colour , shape, size, etc.,

Process: Find the similarity/dissimilarity between objects with features as the defining dimensions

Output:

Group/Cluster of objects



K-Means Clustering.

intra cluster dist $\rightarrow$ Low
inter cluster dist $\rightarrow$ high



x_n μ_k r_{nk}

~~K~~

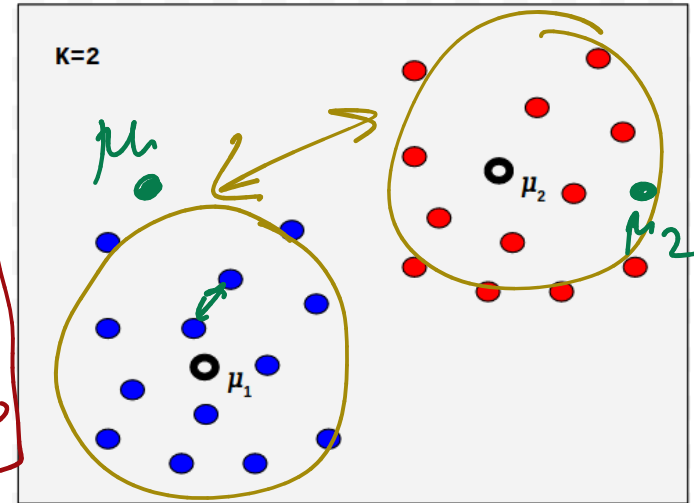
✓ $\{x_1 \dots x_n\}$

K-Means Algorithm

- Works iteratively to find $\{\mu_k\}$ and $\{r_{nk}\}$ such that J is minimized

$$\min J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|x_n - \mu_k\|^2$$

intra



Iteration involves two key steps

Find $\{r_{nk}\}$ fixing $\{\mu_k\}$ to minimize J

Find $\{\mu_k\}$, fixing $\{r_{nk}\}$ to minimize J

Let us look at each of these steps

$x_{21} = \phi$
 $x_{22} = \phi$
 $x_{23} = \phi$

$x_{21} = 1 \rightarrow x_2 \rightarrow C_1$
 $x_{32} = 0 \rightarrow x_3 \neq C_2$

$\mu_k \rightarrow$ representative of C_k
 $\hookrightarrow$ centre

$r_{nk} \rightarrow$ indicator variable $\rightarrow$ point n in cluster k

n point

k clusters.

γ_{nk} = matrix

$$\begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix}$$

$$\begin{aligned} \gamma_{nk} &= 1 \quad \checkmark \\ \uparrow &= 0 \quad \times \end{aligned}$$

$$\begin{array}{cc} n=4 & k=2 \\ & \begin{array}{cc} k_1 & k_2 \end{array} \\ \begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \end{array} & \begin{bmatrix} \gamma_{11} 1 & \gamma_{12} 0 & \dots & \dots & - \\ \gamma_{21} 1 & \gamma_{22} 0 & \dots & \dots & \dots \\ & 0 & \gamma_{32} 1 & & \\ & 1 & & 0 & \end{bmatrix} \end{array}$$

K-Means Algorithm

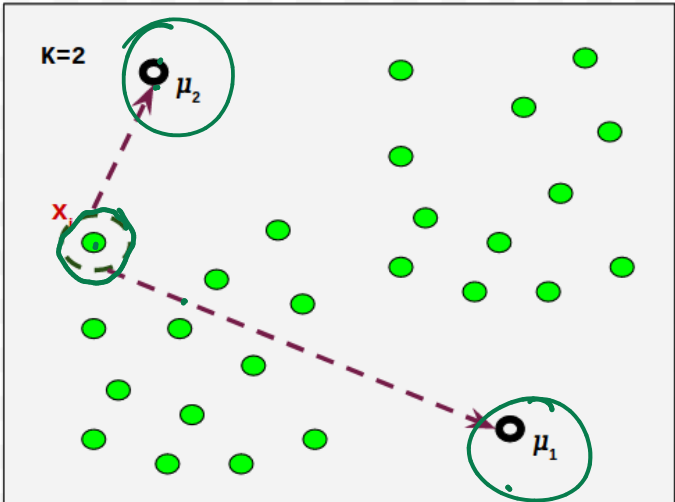
Iter 1 $\rightarrow$ random.

A sample E-Step

① Fix μ_k ??
find x_{nk}
 $\min J$

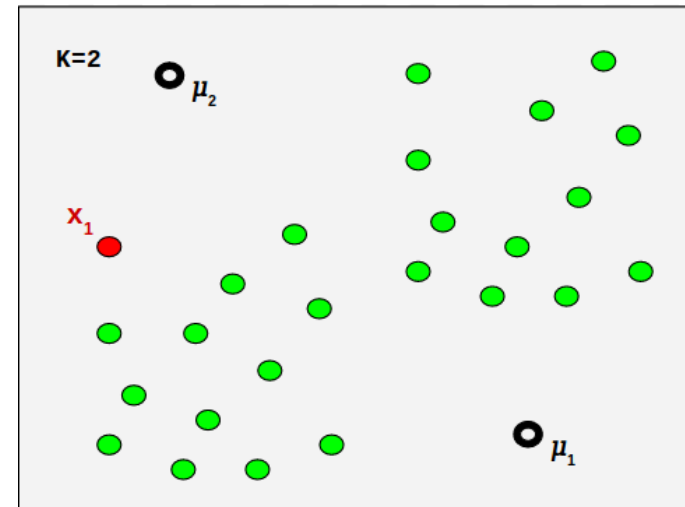
$$\|x_n - \mu_n\|^2$$

$x_i \rightarrow \mu_1$
 $\rightarrow \mu_2$ } smaller ✓



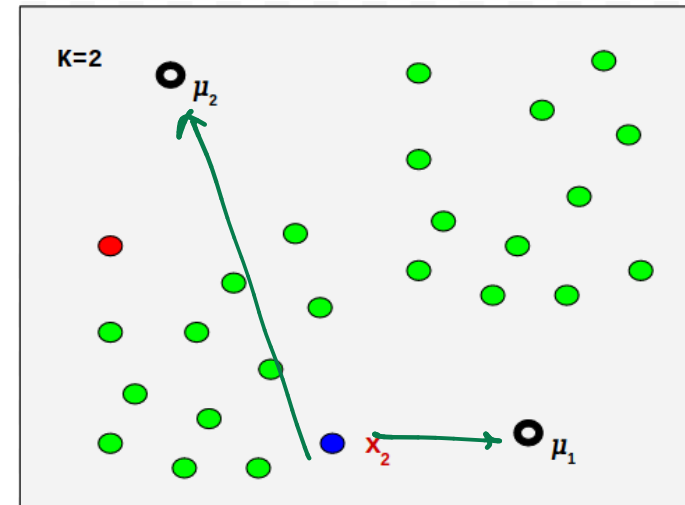
K-Means Algorithm

A sample E-Step



K-Means Algorithm

A sample E-Step



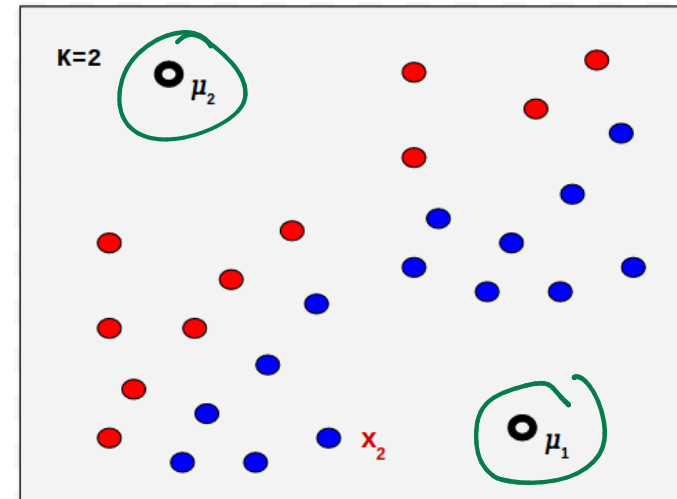
K-Means Algorithm

E-Step:

$$\left[\underset{\uparrow}{r_{nk}} \left\| \underset{\substack{\min \\ \text{min}}}{x_n - \mu_k} \right\|^2 \right] \rightarrow \min J$$

For all $x_t \in X$:

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|x_n - \mu_j\|^2 \\ 0 & \text{otherwise.} \end{cases}$$



K-Means Algorithm

M-Step:

For all μ_k [where $k = 1, 2, \dots, K$]:

$$\mu_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}} \quad \text{mean}$$

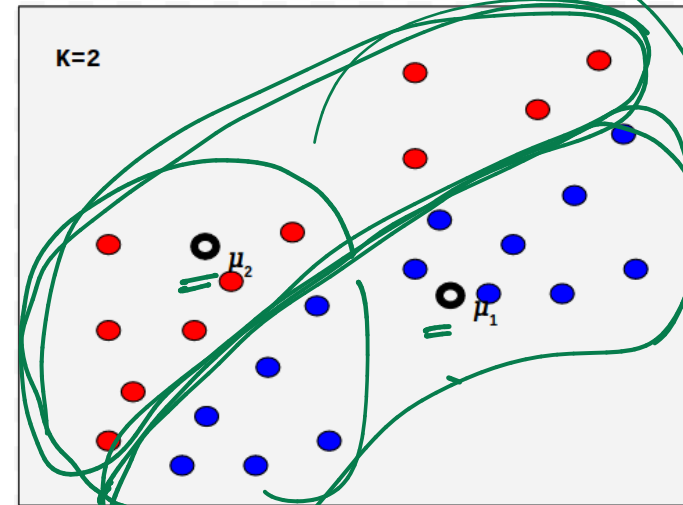
fix r_{nk}

calc μ_k

mi J

$$\frac{1}{\sum} x_i$$

$$10 \rightarrow \frac{\sum x_i}{10}$$



$$r_{nk} ||\mathbf{x}_n - \mu_k||^2$$

K-Means Algorithm

Algorithm:

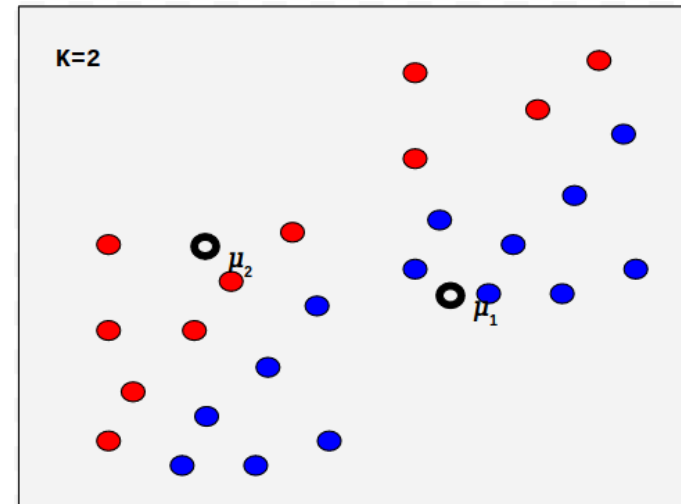
_____ Initialize μ_k [where $k = 1, 2, \dots, K$]

Repeat

E-Step [as defined earlier]

M-Step [as defined earlier]

Until convergence of μ_k .



K-Means Algorithm

Algorithm:

_____ Initialize μ_k [where $k = 1, 2, \dots, K$]

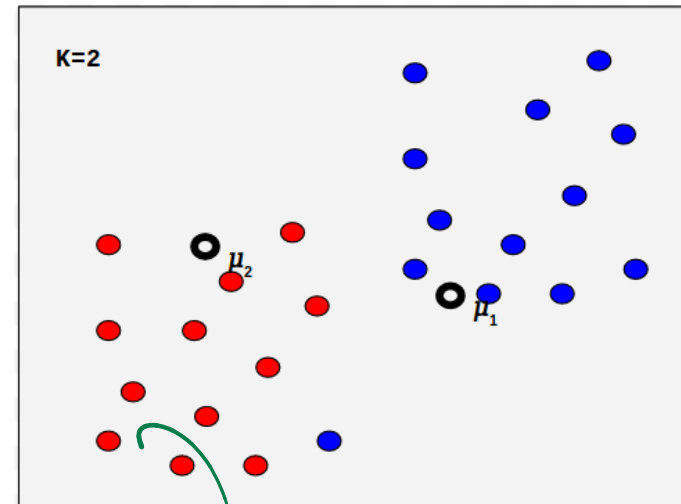
Repeat

E-Step [as defined earlier]

M-Step [as defined earlier]

Until convergence of μ_k .

→ randomly fix μ_k] E₁
 → calc γ_{nk}
 → fix γ_{nk} → calc μ_k] M₁
 → fix μ_k → calc γ_{nk}] E₂
 → γ_{nk} → μ_k] M₂



E-Step in the second iteration

K-Means Algorithm

Algorithm:

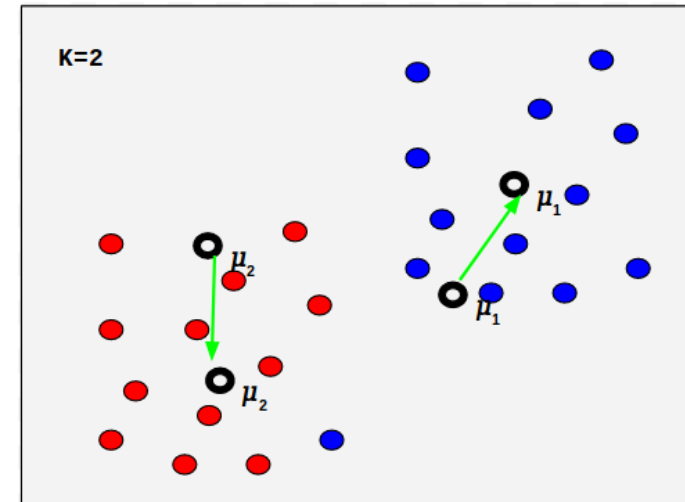
_____ Initialize μ_k [where $k = 1, 2, \dots, K$]

Repeat

E-Step [as defined earlier]

M-Step [as defined earlier]

Until convergence of μ_k .



M-Step in the second iteration

K-Means Algorithm

Algorithm:

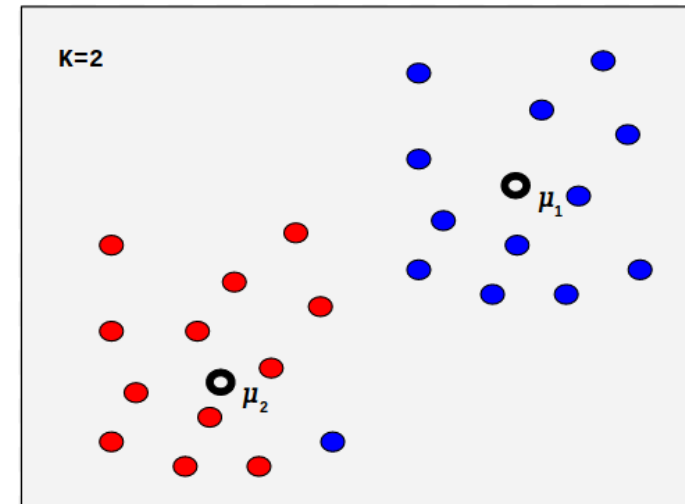
_____ Initialize μ_k [where $k = 1, 2, \dots, K$]

Repeat

E-Step [as defined earlier]

M-Step [as defined earlier]

Until convergence of μ_k .



M-Step in the second iteration

Example

Consider the analysis of weights of individuals and their respective blood glucose levels as given below:

- Identify the clusters using K-means clustering ($k=2$) for the given data, assuming candidate 1 and 2 as initial centroids.
- How many iterations does it take before termination?

Candidate	Weight	Glucose level
1	72	185
2	56	170
3	60	168
4	68	179
5	72	182
6	77	188
7	70	180
8	84	183

Handwritten notes: Yellow arrows point to the 'Weight' and 'Glucose level' headers. Red 'x' and 'y' are written above the headers. Green arrows point from the first two rows to handwritten labels μ_1 and μ_2 .

$$C_1 \rightarrow (1, 4, 5, 6, 7, 8) / k=2$$

Example

$$\|x_k - \mu_k\|^2$$

$$5^x$$

$$\gamma_{1,1} = 1$$

$$\gamma_{nk}$$

Candidate	Cluster1(72, 185)	Cluster2(56, 170)
3(50, 168)	= 17 ² + 12 ²	= 2 ² + 4 ²
4(68, 179)	= 6 ² + 4 ²	= 9 ² + 12 ²
5(72, 182)	= 3 ² + 0 ²	= 12 ² + 16 ²
6(77, 188)	= 3 ² + 5 ²	= 18 ² + 11 ²
7(70, 180)	= 5 ² + 2 ²	= 10 ² + 14 ²
8(183, 84)	= 2 ² + 12 ²	= 13 ² + 28 ²

After 1st iteration, the cluster groups are C1{1, 4, 5, 6, 7, 8} and C2{2, 3}

Re-computing centroids:

$$C1([72+68+72+77+70+84]/6, [185+179+182+188+180+183]/6) = C1(73.83, 182.83)$$

$$C2([56+60]/2, [170+168]/2) = C2(58, 169)$$

2nd Iteration –computing distances from resulting centroids:

After 2nd iteration, the cluster groups are C1{1, 4, 5, 6, 7, 8} and C2{2, 3}; there is no change in cluster groups hence the algorithm terminates.

Candidate	Cluster1(182.83, 73.83)	Cluster2(169, 58)
1(185, 72)	(182.83 - 185) ² + (73.83 - 72) ² = 2.17 ² + 1.83 ²	(169 - 185) ² + (58 - 72) ² = 16 ² + 14 ²
2(170, 56)	(182.83 - 170) ² + (73.83 - 56) ² = 12.83 ² + 17.83 ²	(169 - 170) ² + (58 - 56) ² = 1 ² + 2 ²
3(168, 60)	(182.83 - 168) ² + (73.83 - 60) ² = 14.83 ² + 13.83 ²	(169 - 168) ² + (58 - 60) ² = 1 ² + 2 ²
4(179, 68)	(182.83 - 179) ² + (73.83 - 68) ² = 3.83 ² + 5.83 ²	(169 - 179) ² + (58 - 68) ² = 10 ² + 10 ²
5(182, 72)	(182.83 - 182) ² + (73.83 - 72) ² = 0.83 ² + 1.83 ²	(169 - 182) ² + (58 - 72) ² = 13 ² + 14 ²
6(188, 77)	(182.83 - 188) ² + (73.83 - 77) ² = 5.17 ² + 3.17 ²	(169 - 188) ² + (58 - 77) ² = 19 ² + 19 ²
7(180, 70)	(182.83 - 180) ² + (73.83 - 70) ² = 2.17 ² + 1.83 ²	(169 - 180) ² + (58 - 70) ² = 11 ² + 12 ²
8(183, 84)	(182.83 - 183) ² + (73.83 - 84) ² = 0.17 ² + 10.17 ²	(169 - 183) ² + (58 - 84) ² = 14 ² + 26 ²

SSE + k

Evaluating K-means Clusters

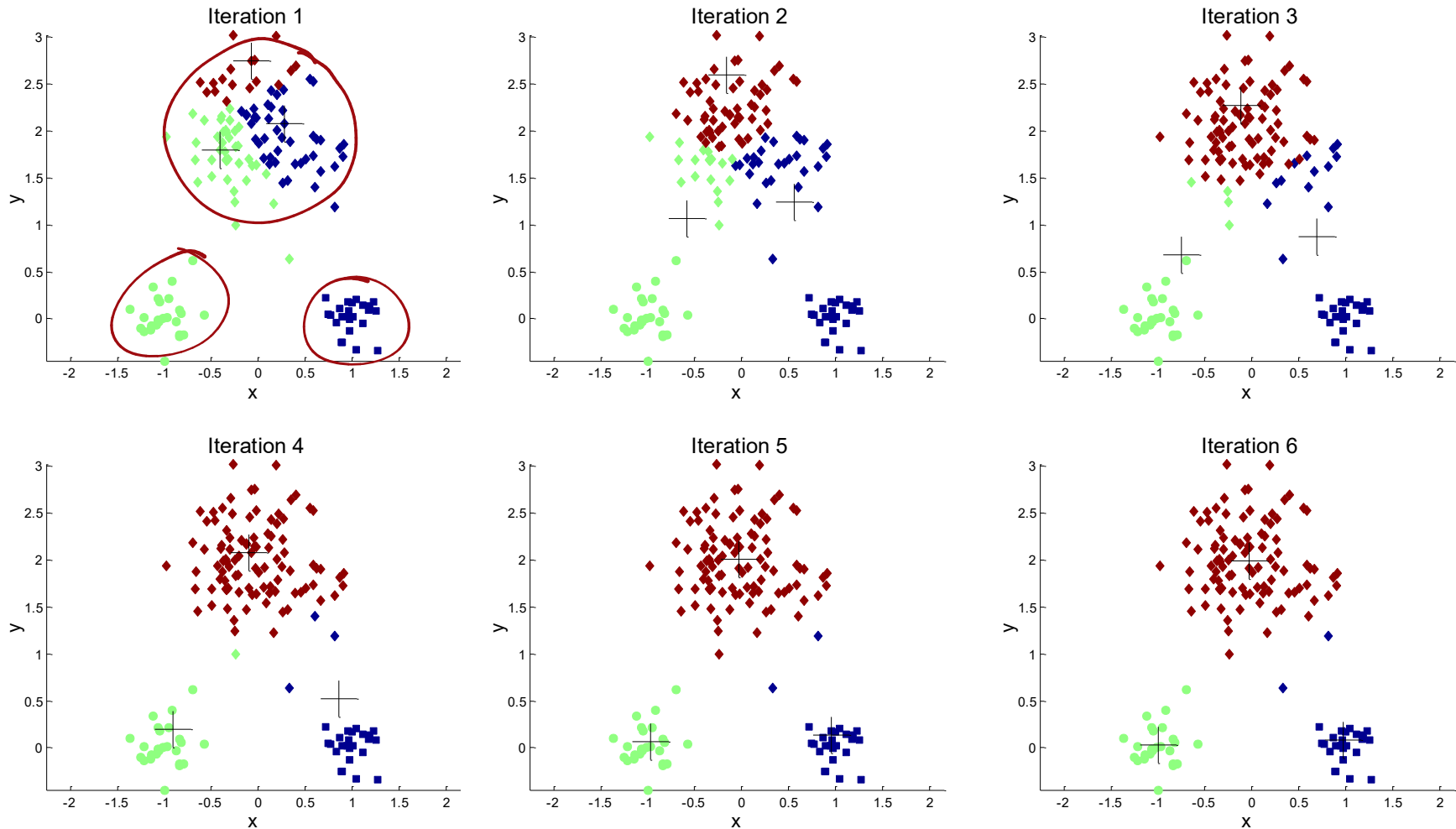
- Most common measure is Sum of Squared Error (SSE)
 - For each point, the error is the distance to the nearest cluster
 - To get SSE, we square these errors and sum them.

→ k = 2
k ↑ SSE ↓

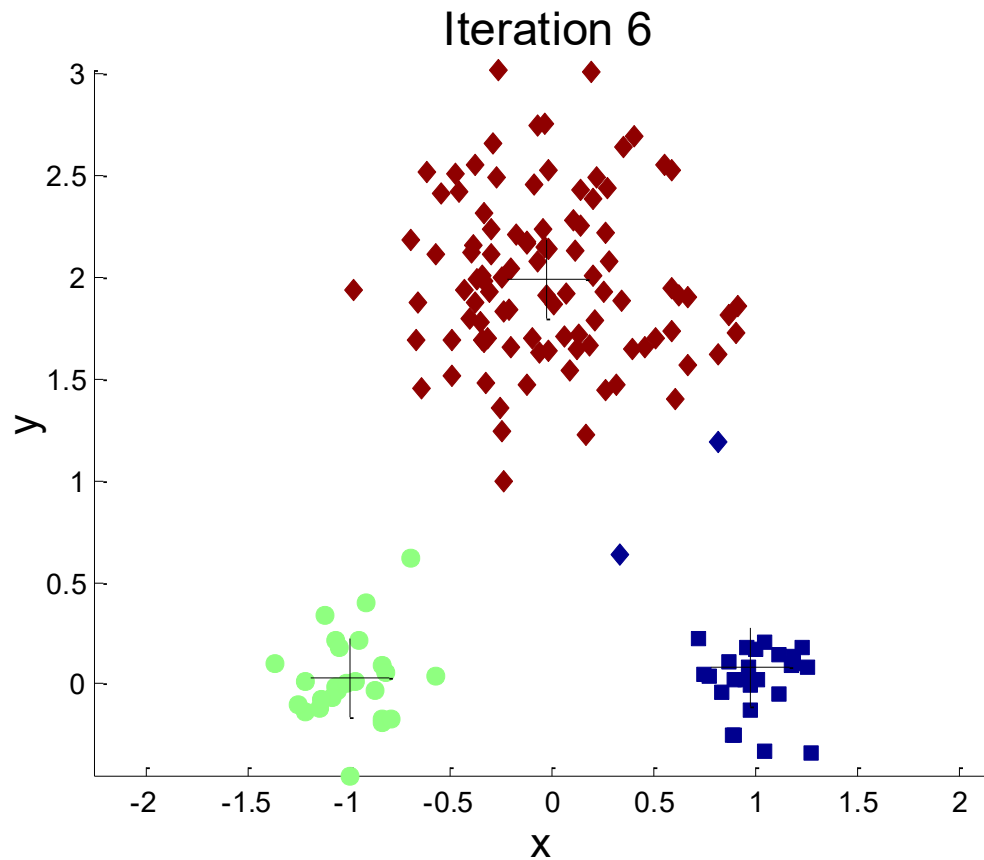
$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- x is a data point in cluster C_i and m_i is the representative point for cluster C_i
 - can show that m_i corresponds to the center (mean) of the cluster
- Given two clusters, we can choose the one with the smallest error
- One easy way to reduce SSE is to increase K , the number of clusters
 - A good clustering with smaller K can have a lower SSE than a poor clustering with higher K

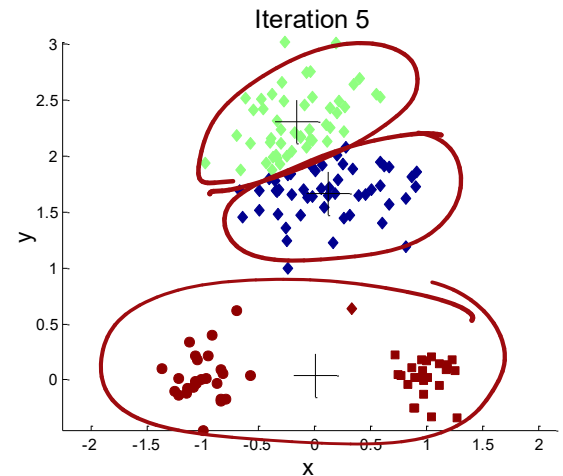
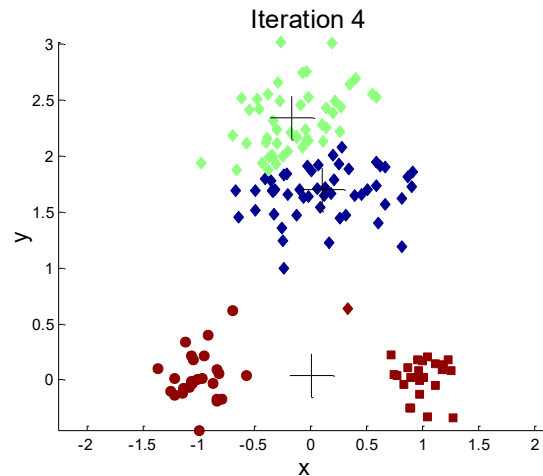
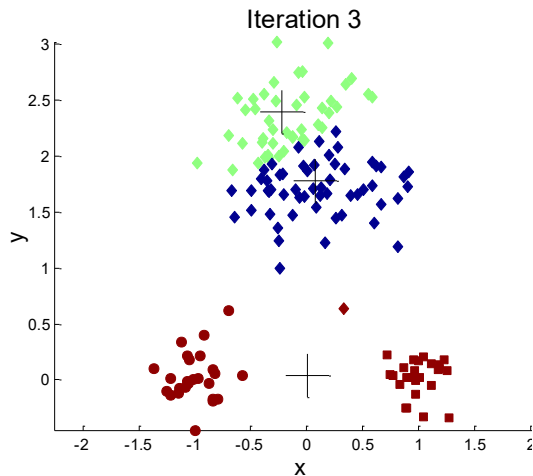
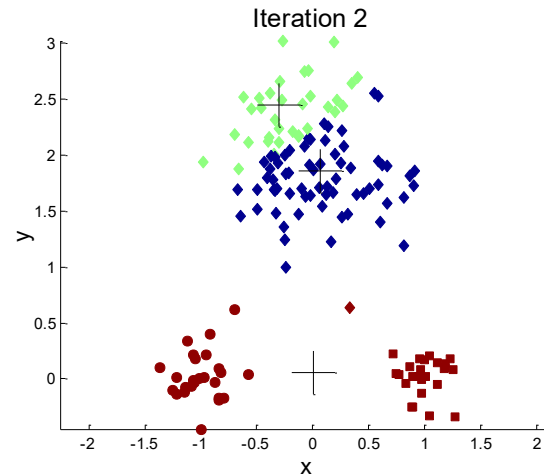
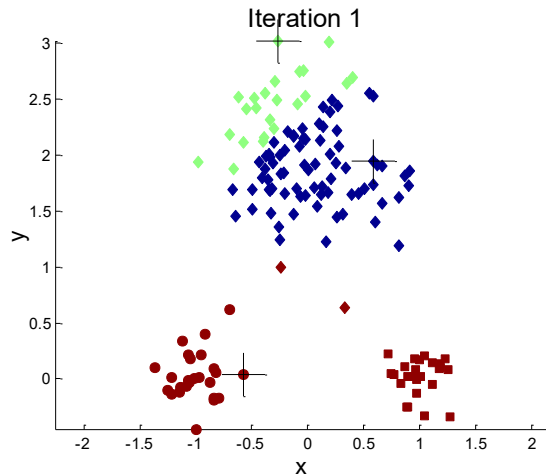
Importance of Choosing Initial Centroids

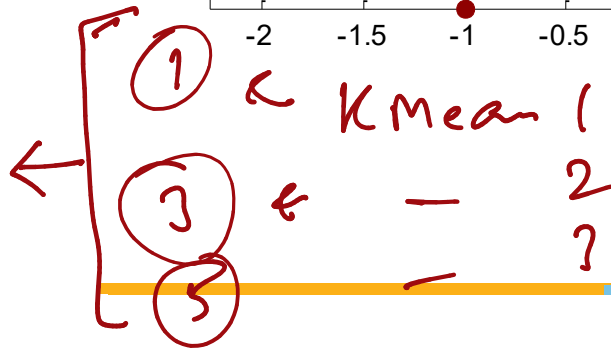


Importance of Choosing Initial Centroids



Importance of Choosing Initial Centroids ...

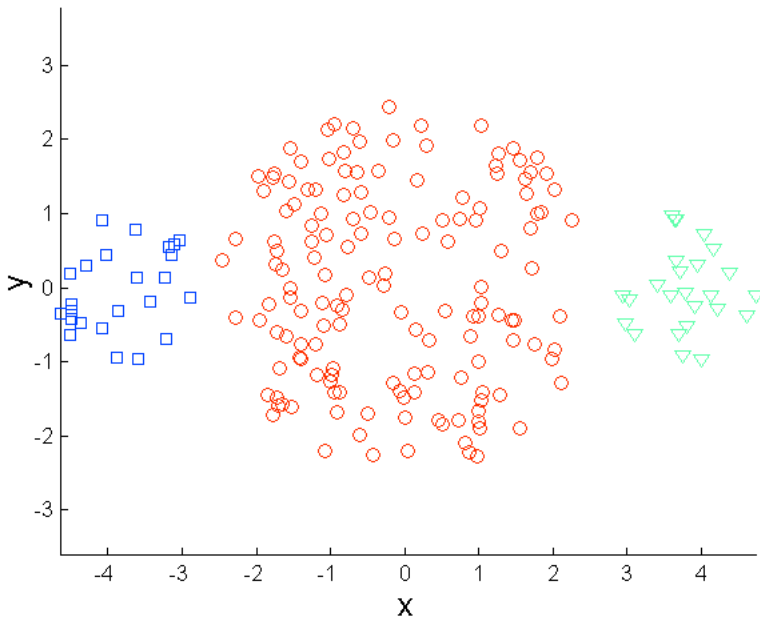




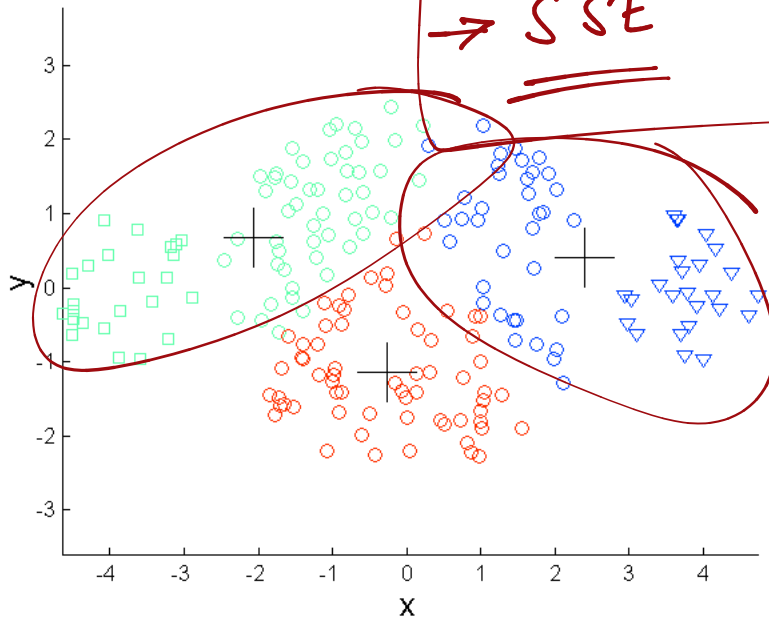
Limitations of K-means: Differing Sizes

prefer-
→ min. size
spread

→ SSE

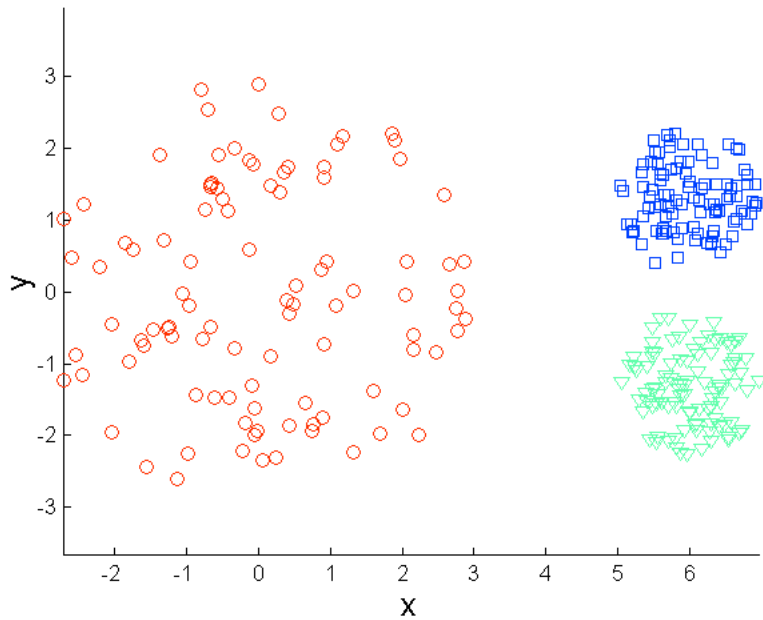


Original Points

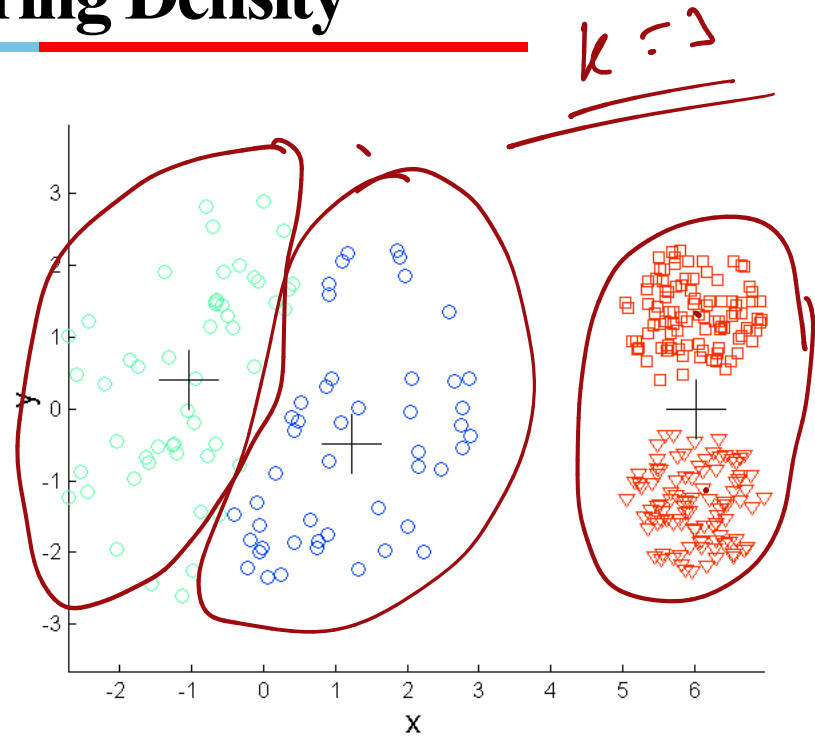


K-means (3 Clusters)

Limitations of K-means: Differing Density



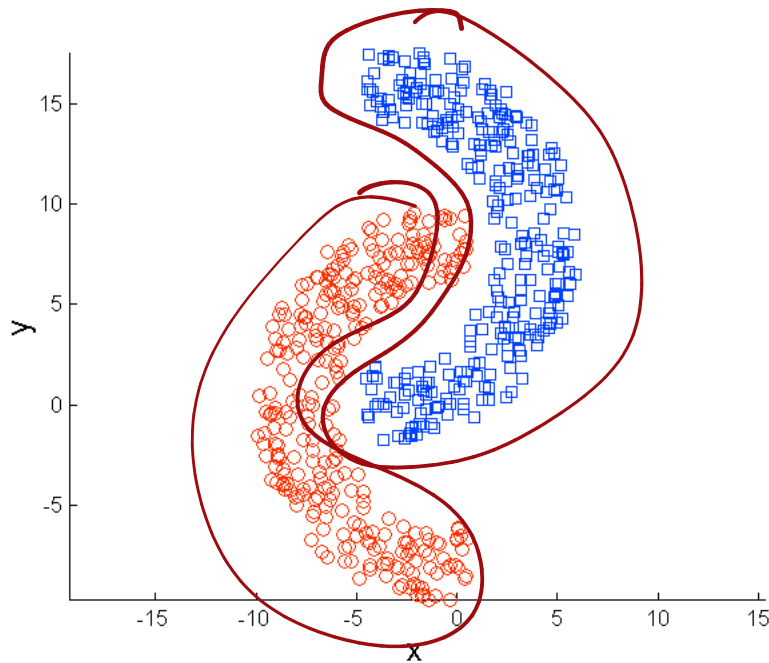
Original Points



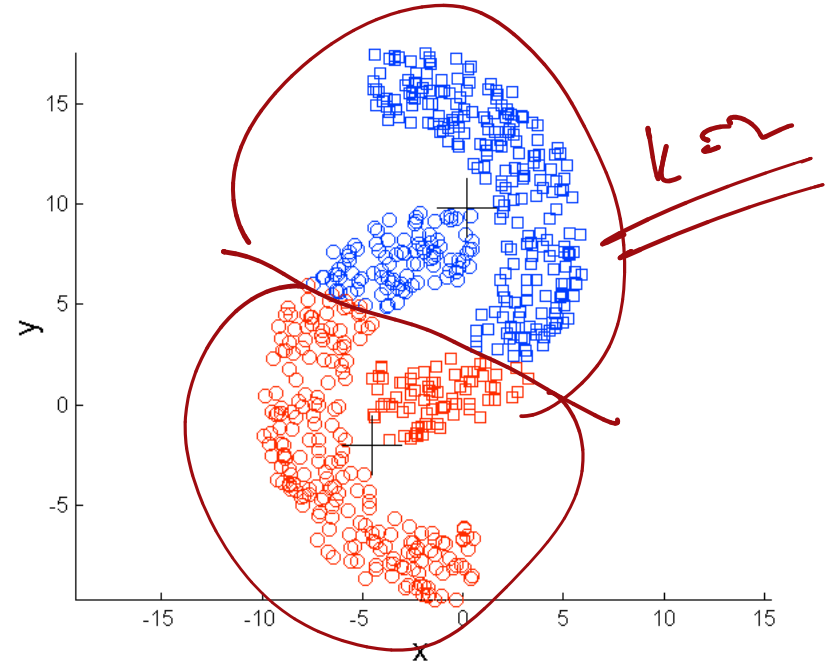
K-means (3 Clusters)



Limitations of K-means: Non-globular Shapes



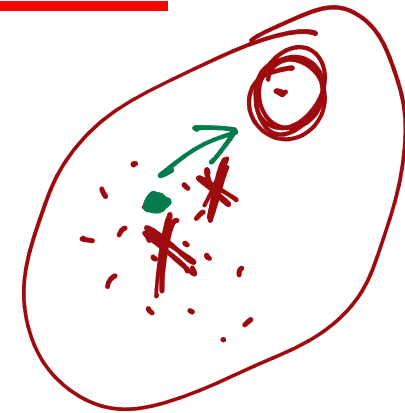
Original Points



K-means (2 Clusters)

Limitations of K-means

- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes
- K-means has problems when the data contains outliers.



$$M \rightarrow \mu_k$$

mean

Sol?



From K-Means to K-Medoids

mean
↑

median
↑

- The k-means is sensitive to outliers-since an object with an extremely large value may substantially distort the distribution of the data
- K-Medoids: Instead of taking the mean value of the object in a cluster as a reference point, medoids can be used, which is the most centrally located object in a cluster
- **K-Medoids**

→ part of my dataset

1. Initialize: select k random points out of the n data points as the medoids.
2. Associate each data point to the closest medoid by using any common distance metric methods.
3. While the cost decreases:
For each medoid m , for each data point o which is not a medoid:
 1. Swap m and o , associate each data point to the closest medoid, recompute the cost.
 2. If the total cost is more than that in the previous step, undo the swap.

How to choose a “good” k for k-means clustering?

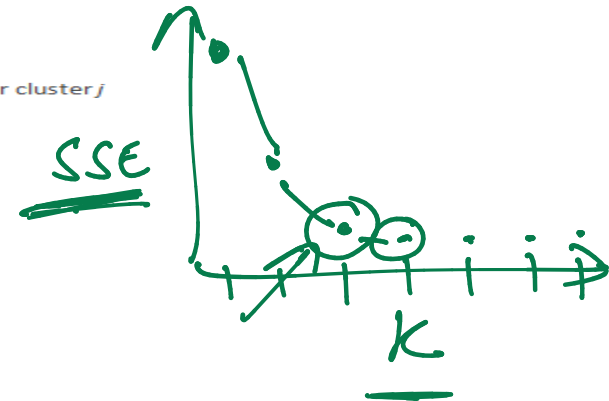


- Choose the number of clusters by visually inspecting data points
- **Elbow method:**
 - Compute the sum of squared error (SSE) for some values of k (for example, 2,3,4,5,6...). The SSE is defined as the sum of the squared distance between each member of the cluster and its centroid.

$$\text{objective function} \leftarrow J = \sum_{j=1}^k \sum_{i=1}^n \|x_i^{(j)} - c_j\|^2$$

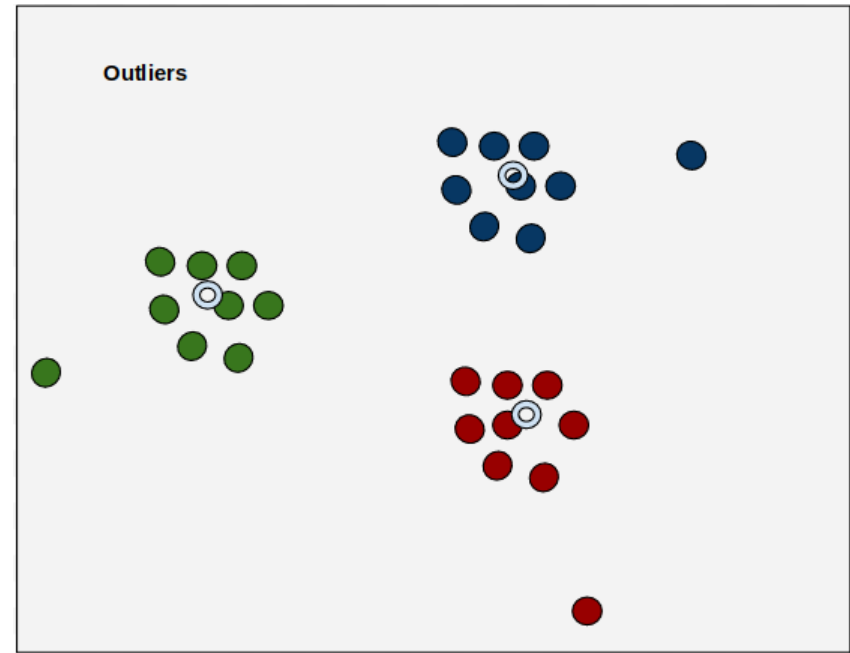
Diagram illustrating the formula for the objective function (SSE):

- k : number of clusters
- n : number of cases
- $x_i^{(j)}$: case i
- c_j : centroid for cluster j
- $\|x_i^{(j)} - c_j\|^2$: Distance function



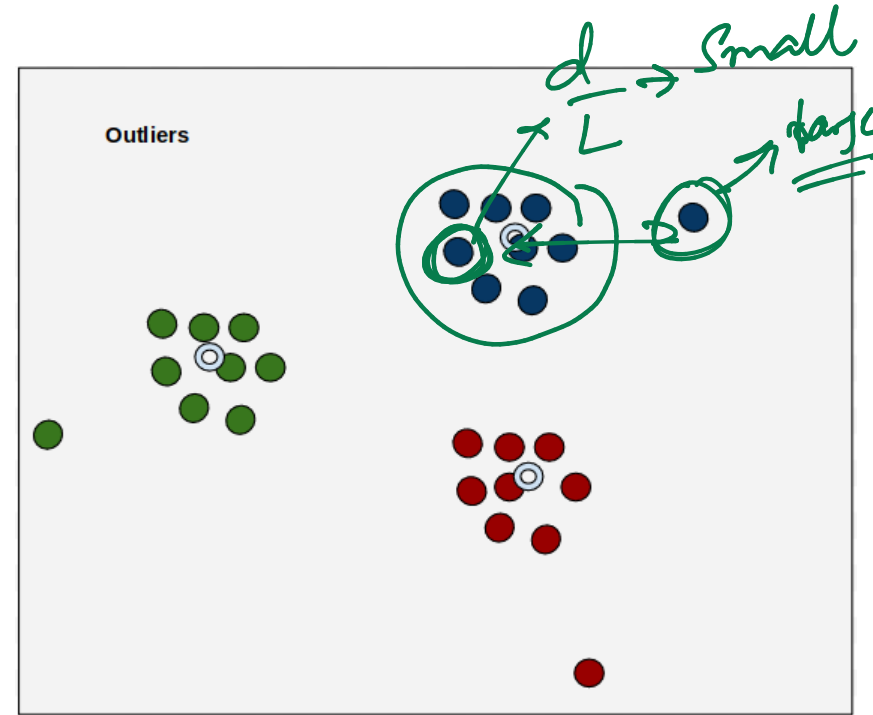
- If you plot k against SSE, you will see that the error decreases as k gets larger, this is because when the number of clusters increases, they should be smaller, so distortion is also smaller. The idea of the elbow method is to choose the k at which the SSE decreases abruptly.

(naive) K-Means for detecting outliers



(naive) K-Means for detecting outliers

- Let
- $\text{dist}(x, \mu_k)$ be the distance of a point x , assigned to cluster k to its center μ_k .
- L_{μ_k} be the average distance of all the points assigned to cluster k with its center
- The ratio $\left[\frac{\text{dist}(x, \mu_k)}{L_{\mu_k}} \right]$ for each point is the outlier score for each point.
- Higher the ratio for a point x , more likely x is an outlier



$$\left(\gamma_{nk} \frac{\|x_n - \mu_k\|^2}{L_{\mu_k}} \right)$$

Handwritten annotations: x_n points to x_n , μ_k points to μ_k , and L_{μ_k} points to L_{μ_k} .

GMM

Notion of Mixture Models

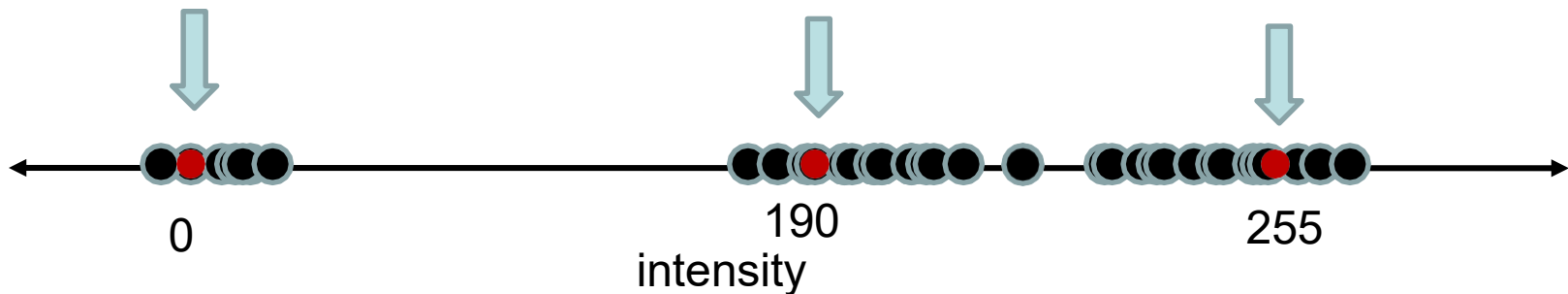
K means – Hard clustering

- Assigned each example to exactly one cluster
- What if clusters are overlapping?
 - Hard to tell which cluster is right
 - Maybe we should try to remain uncertain
- What if cluster has a non-circular shape?
- Hard assignments of data points to clusters – small shift of a data point can flip it to a different cluster
- For example, if a point is near the 'border' between two clusters, it's often better to know that it has near equal membership probabilities for these clusters, rather than blindly assigning it to the nearest one.

$$x_{nk} = 1, 0$$

hard

NOT OVERLAP



1 of K coding mechanism

- For each data point x_n , we introduce a set of **binary indicator variables**

$r_{nk} \in [0,1]$ such that

- $$\sum_k r_{nk} = 1$$

where $k = 1, \dots, K$ describing which of the K clusters the data point x_n is assigned to, so that if data point x_n is assigned to cluster k then $r_{nk} > r_{nj}$ for j not equal to k .

- Example: 5 data points and 3 clusters

$$(r_{nk}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Handwritten notes:

$x_1 \rightarrow \begin{matrix} C_1 \\ 0.8 \end{matrix} \quad \begin{matrix} C_2 \\ 0.15 \end{matrix} \quad \begin{matrix} C_3 \\ 0.05 \end{matrix} \Bigg\} = 1$

$\leftarrow$ hard

GMM - soft Clustering

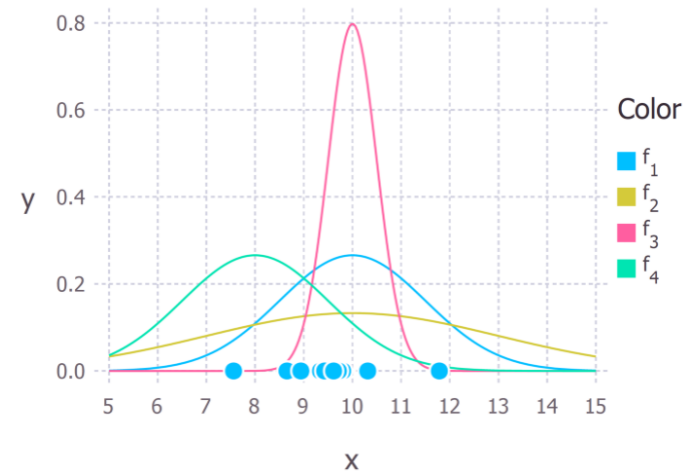
- **Solution:** replace 'hard' clustering of K-means with 'soft' probabilistic assignments ✓
- Probabilistic Clustering
 - Represents the probability distribution of the data as a **Gaussian mixture model (GMM)**
 - GMMs give a probabilistic assignment of points to clusters which quantify uncertainty.
 - Mixture models combines a set of distributions to create a convex space where we can search for the optimal parameters for such distributions using MLE.
- In GMM, Clusters modeled as Gaussians
- **EM algorithm:** assign data to cluster with some probability

Source Credit: Christopher M. Bishop

Gaussian Distribution

- The normal curve is *bell-shaped* and has a single peak at the
- exact center of the distribution.
- The arithmetic mean, median, and mode of the distribution are equal and located at the peak.
- Half the area under the curve is above the peak, and the other half is below it.
- The normal distribution is symmetrical about its mean.

$\left[\begin{array}{l} \mu \sim \\ \sigma \sim \end{array} \right]$



The 10 data points and possible Gaussian distributions from which the data were drawn. f_1 is normally distributed with mean 10 and variance 2.25 (variance is equal to the square of the standard deviation), this is also denoted $f_1 \sim N(10, 2.25)$. $f_2 \sim N(10, 9)$, $f_3 \sim N(10, 0.25)$ and $f_4 \sim N(8, 2.25)$.

MATHEMATICAL FUNCTION (Pdf)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Note constants:

$\pi=3.14159$

$e=2.71828$

This is a bell shaped curve with different centers and spreads depending on μ and σ

Univariate and Multivariate Gaussian density

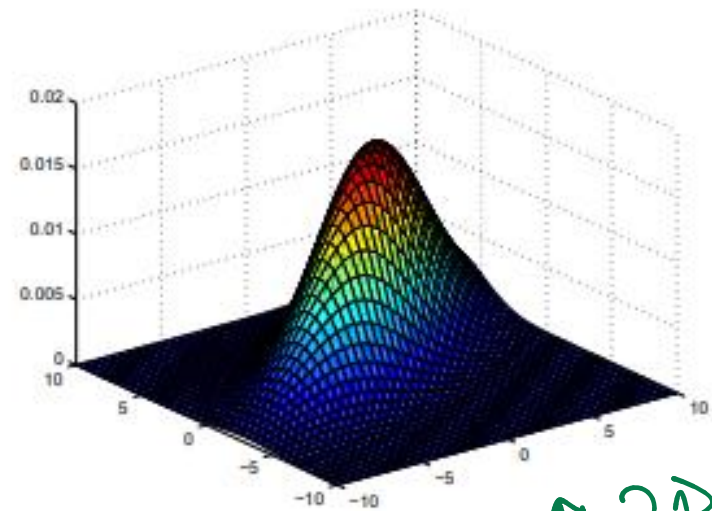
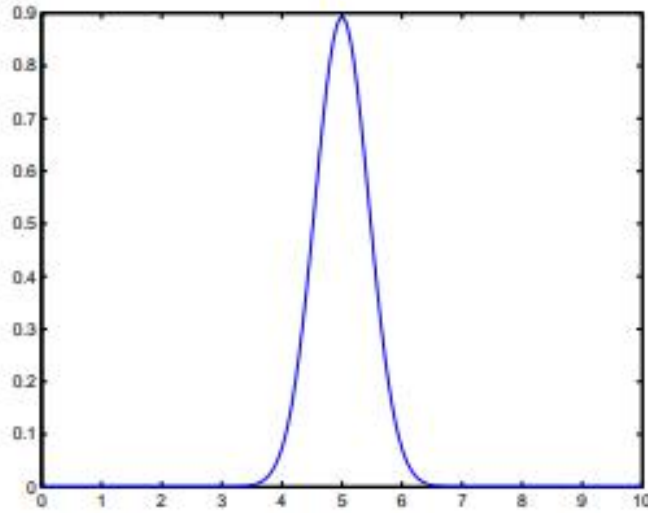
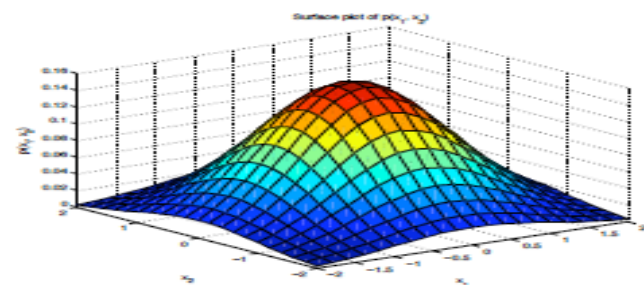


Figure 1: The figure on the left shows a univariate Gaussian density for a single variable X . The figure on the right shows a multivariate Gaussian density over two variables X_1 and X_2 .

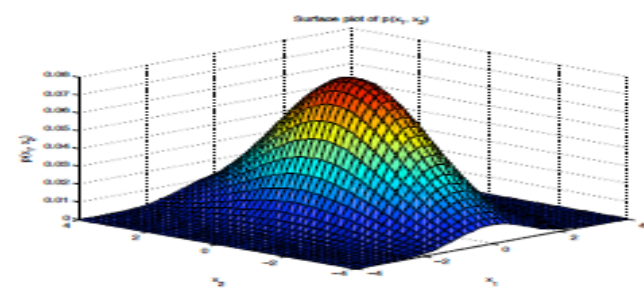
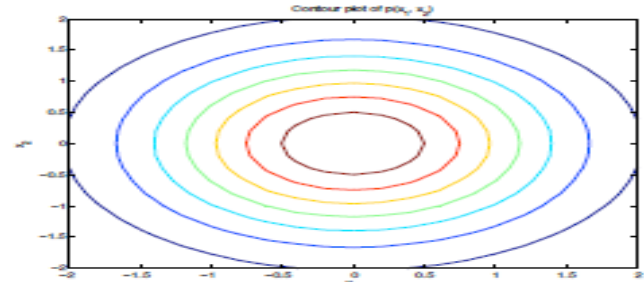
$x \rightarrow$ feature

$\sigma^2 \rightarrow \Sigma$

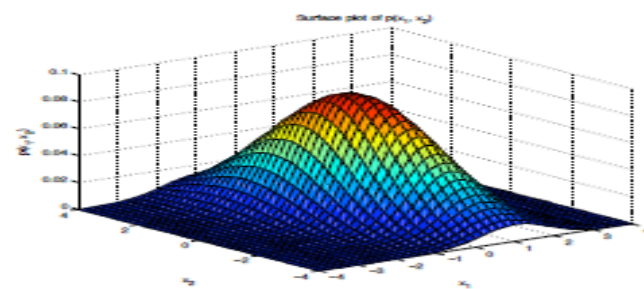
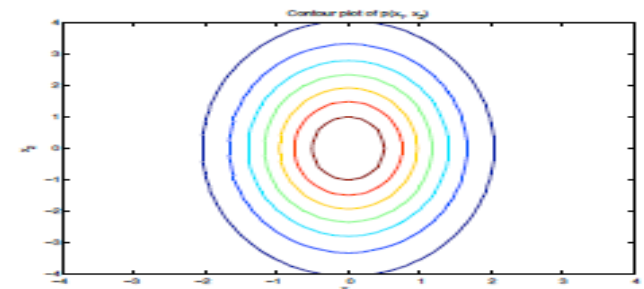
2-dimensional Gaussian distribution



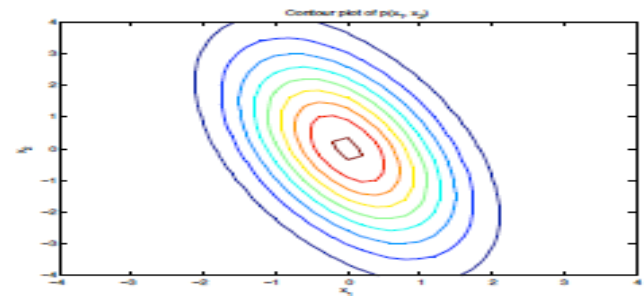
(a) Spherical Gaussian (diagonal covariance, equal variances)



(b) Gaussian with diagonal covariance matrix



(c) Gaussian with full covariance matrix



$\rightarrow \mu, \Sigma$
 $\uparrow \quad \uparrow$
peak spread
 $\Sigma = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$
 $\uparrow$
 $\Sigma = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$
 $\uparrow$
 $\Sigma = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

Figure 7: Surface and contour plots of 2-dimensional Gaussian with different covariance structures

k means vs GMM

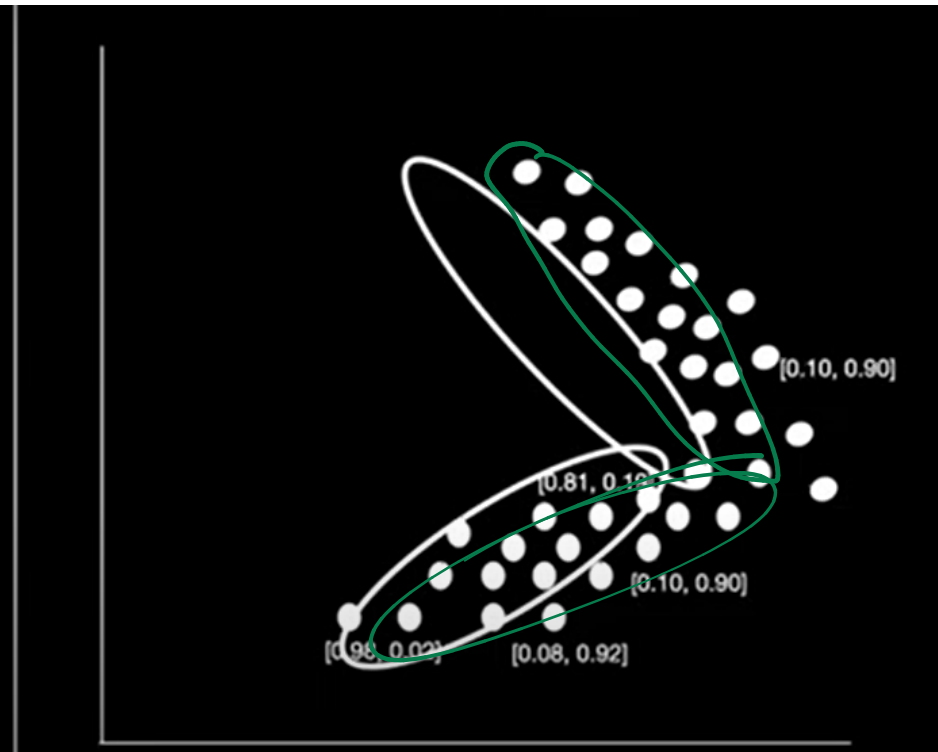
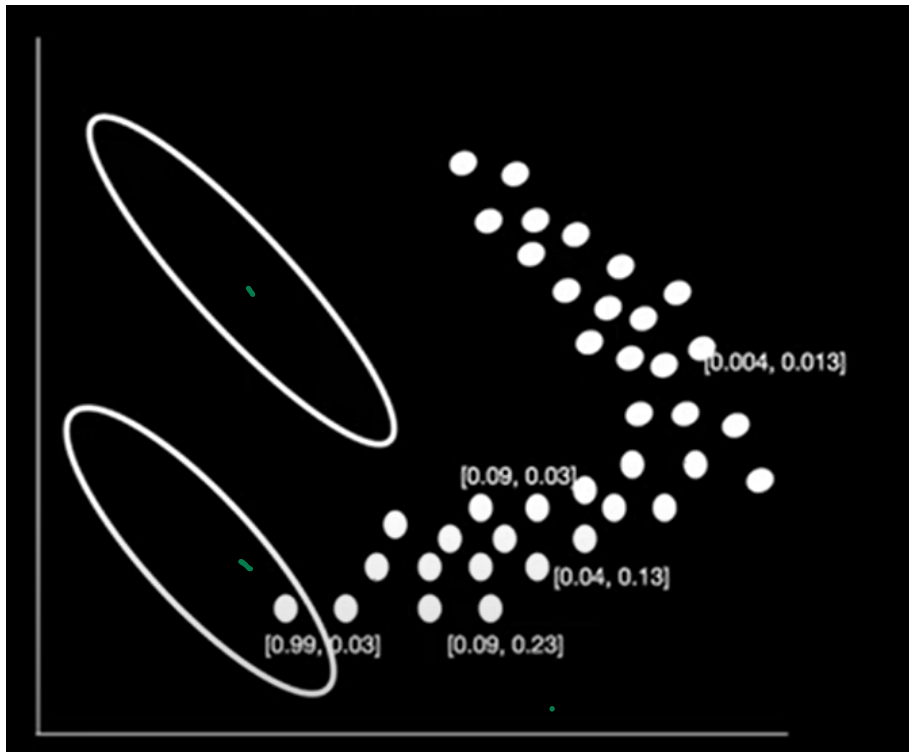


$k = 2$

(μ_1, Σ_1) (μ_2, Σ_2)

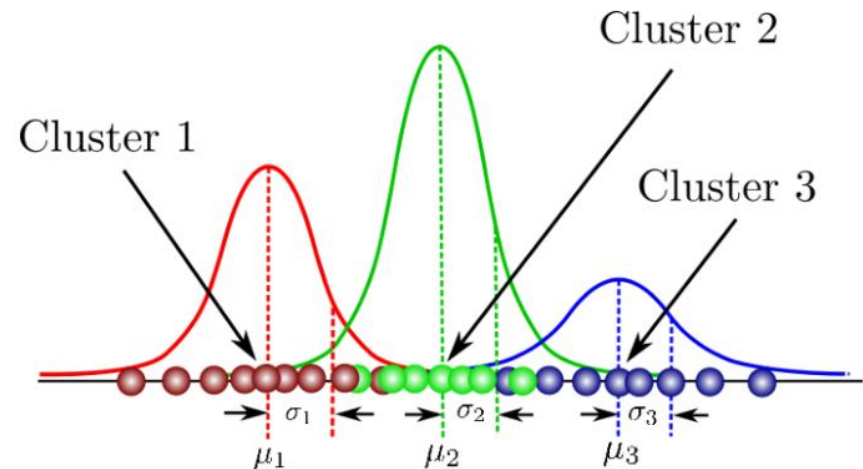
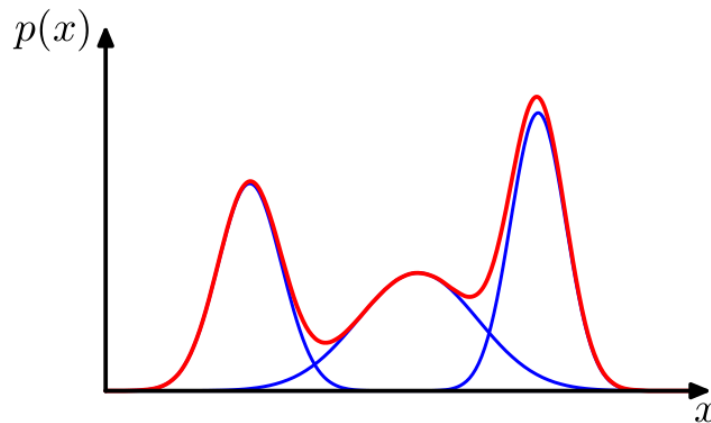
Expectation Step
Assignment

Maximization Step
Update



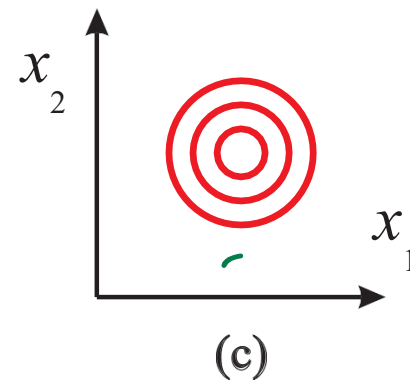
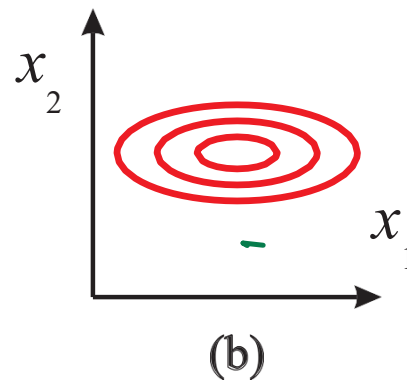
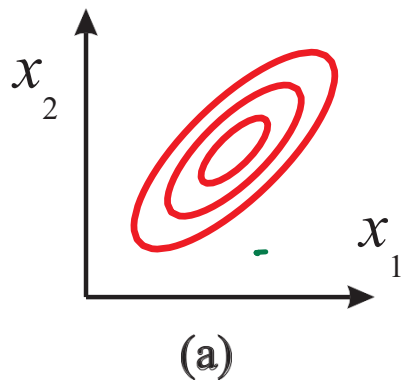
GMM – Mixture of Gaussians

- Probabilistic cluster is a distribution over the data space, which can be mathematically represented using a probability density function (or distribution function).
- Our task: infer a set of k probabilistic clusters that is mostly likely to generate D



Example: Mixture of 3 Gaussians

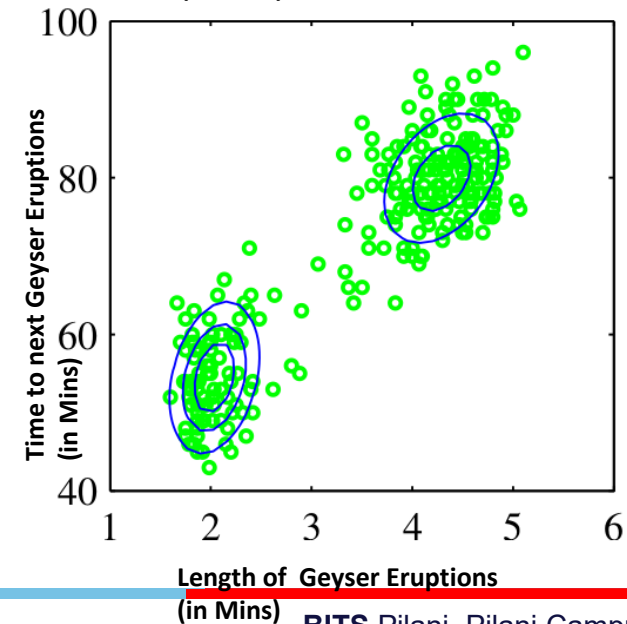
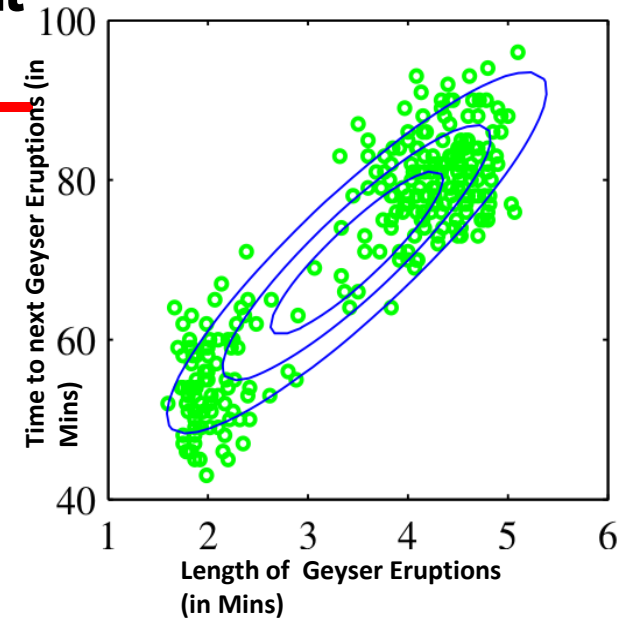
$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$



- The observed data come from a Gaussian mixture distribution which consists of K Gaussians with their own means and variances.
- K classes are latent

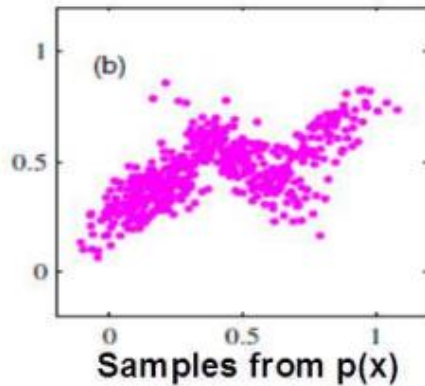
GMM – Why single distribution is insufficient

- Gaussian distribution is intrinsically unimodal
- A single distribution is not sufficient to fit many real data sets which are multimodal
- A mixture model assumes that a set of observed objects is a mixture of instances from multiple probabilistic clusters, and conceptually each observed object is generated independently

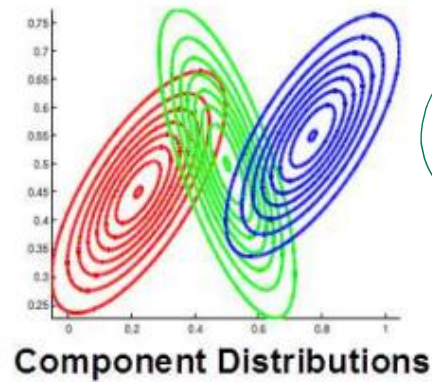


Learning a Mixture of Gaussians

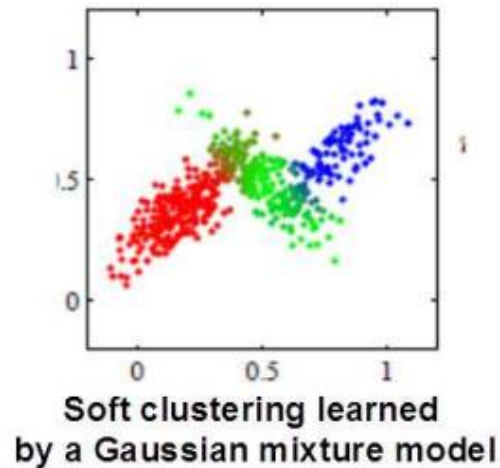
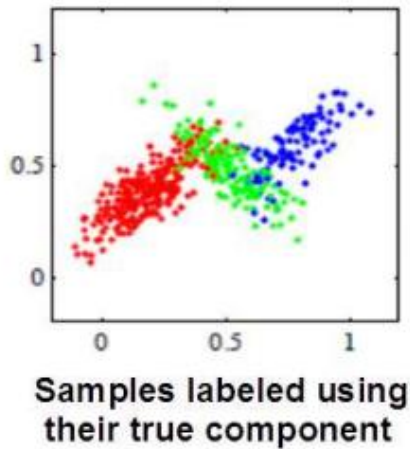
Our actual observations



Mixture of 3 Gaussians

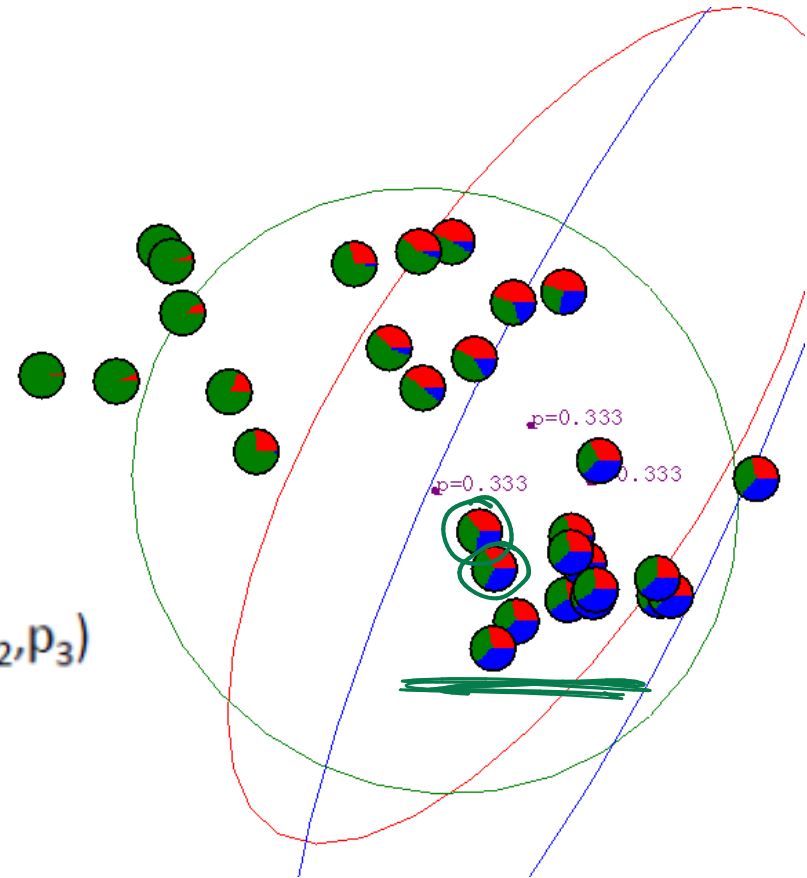
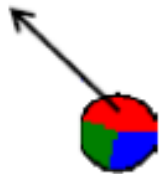


$k = 3$

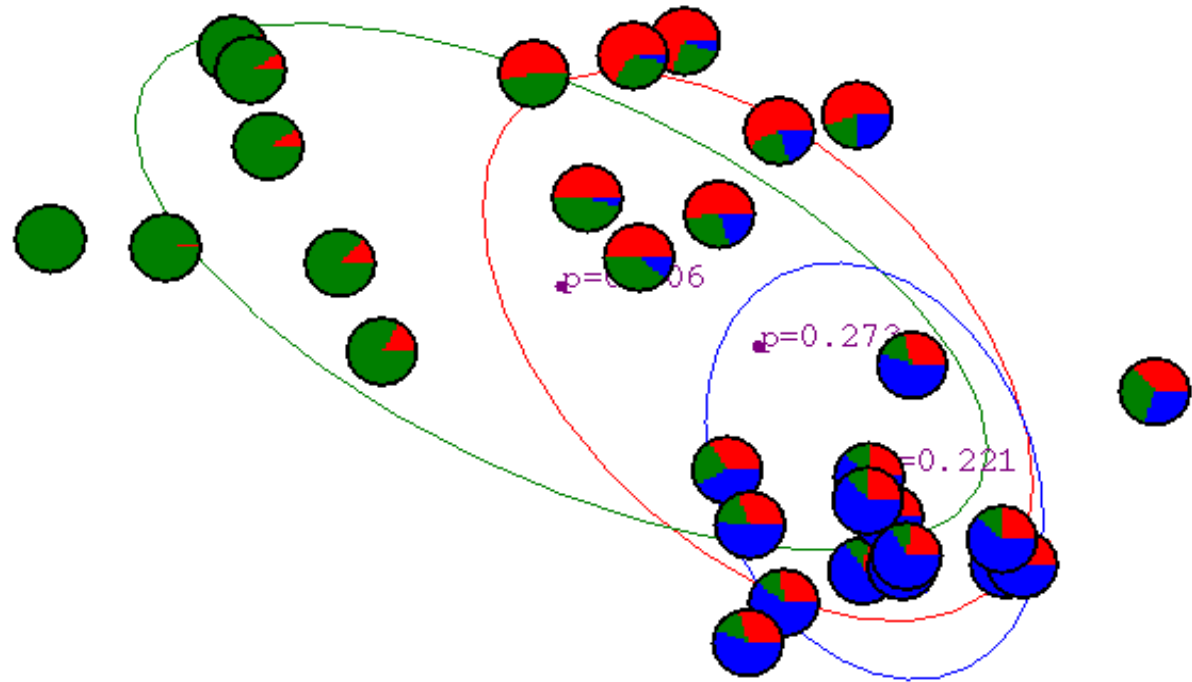


Gaussian Mixture Example: Start

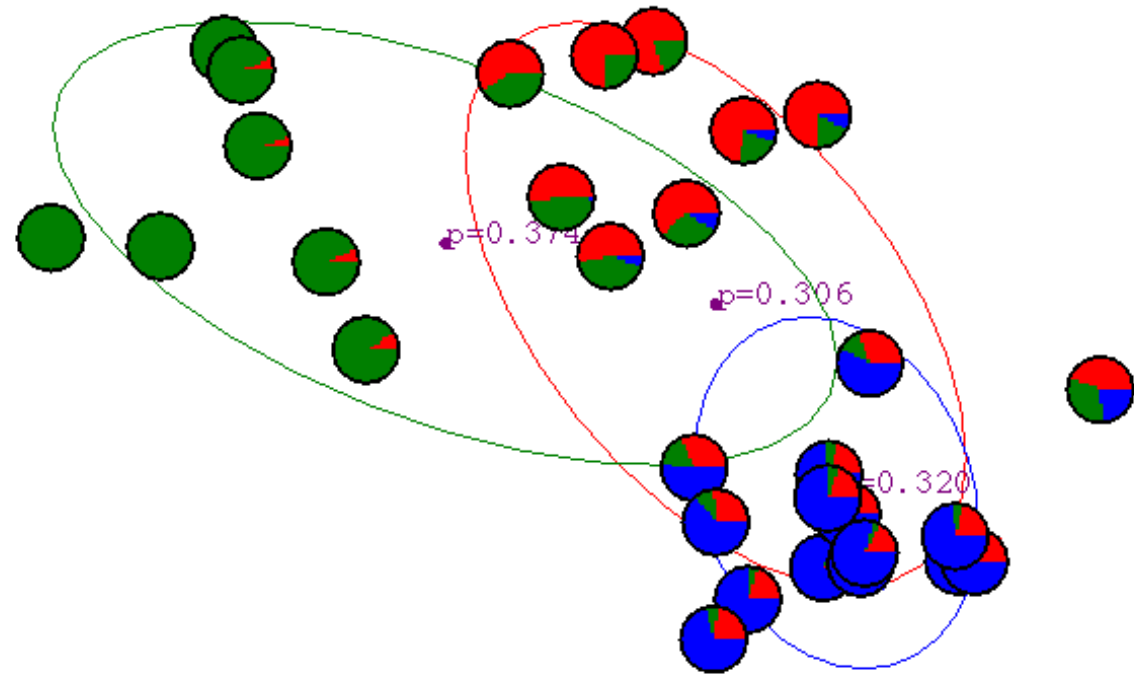
$$P(y = \bullet | x_j, \mu_1, \mu_2, \mu_3, \Sigma_1, \Sigma_2, \Sigma_3, p_1, p_2, p_3)$$



After first iteration

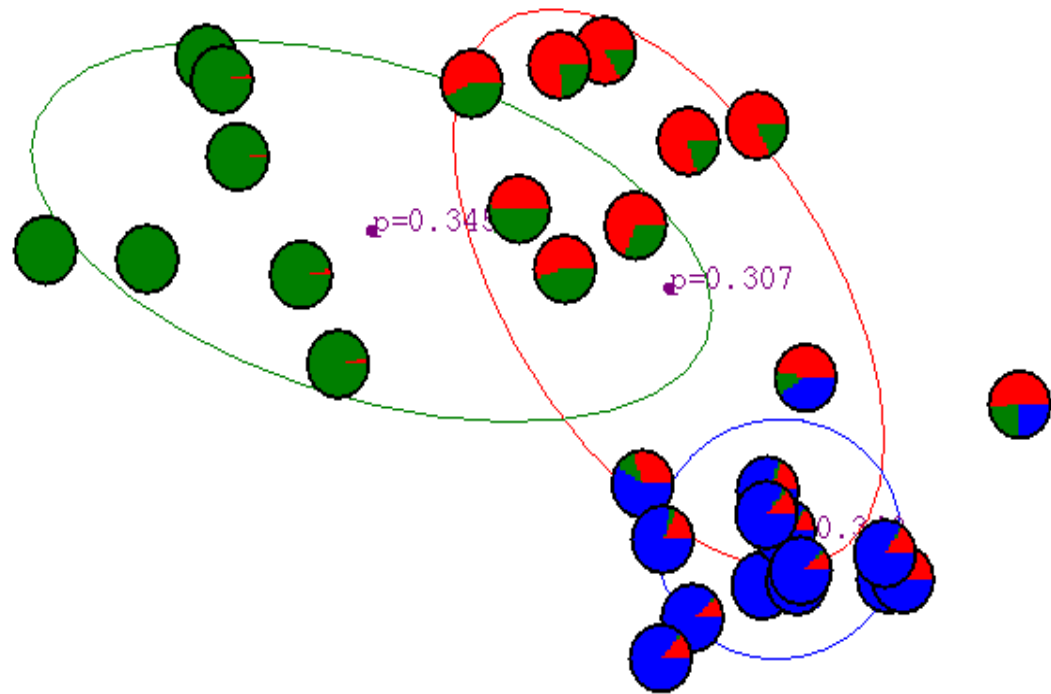


After 2nd iteration



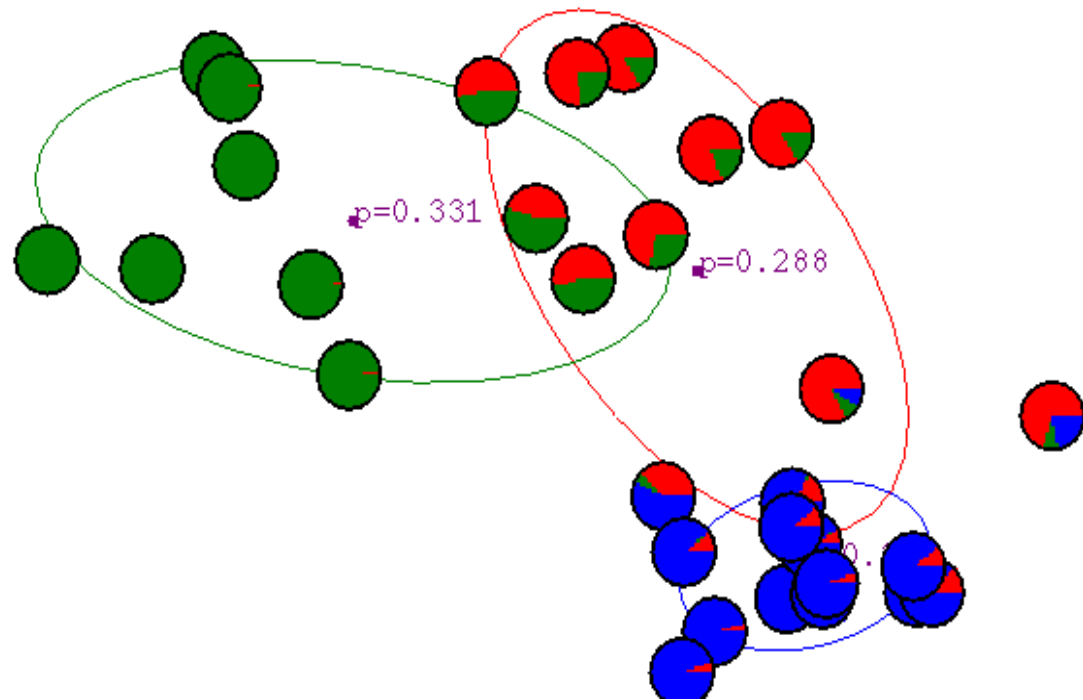
Slide Credit: Andrew W. Moore

After 3rd iteration

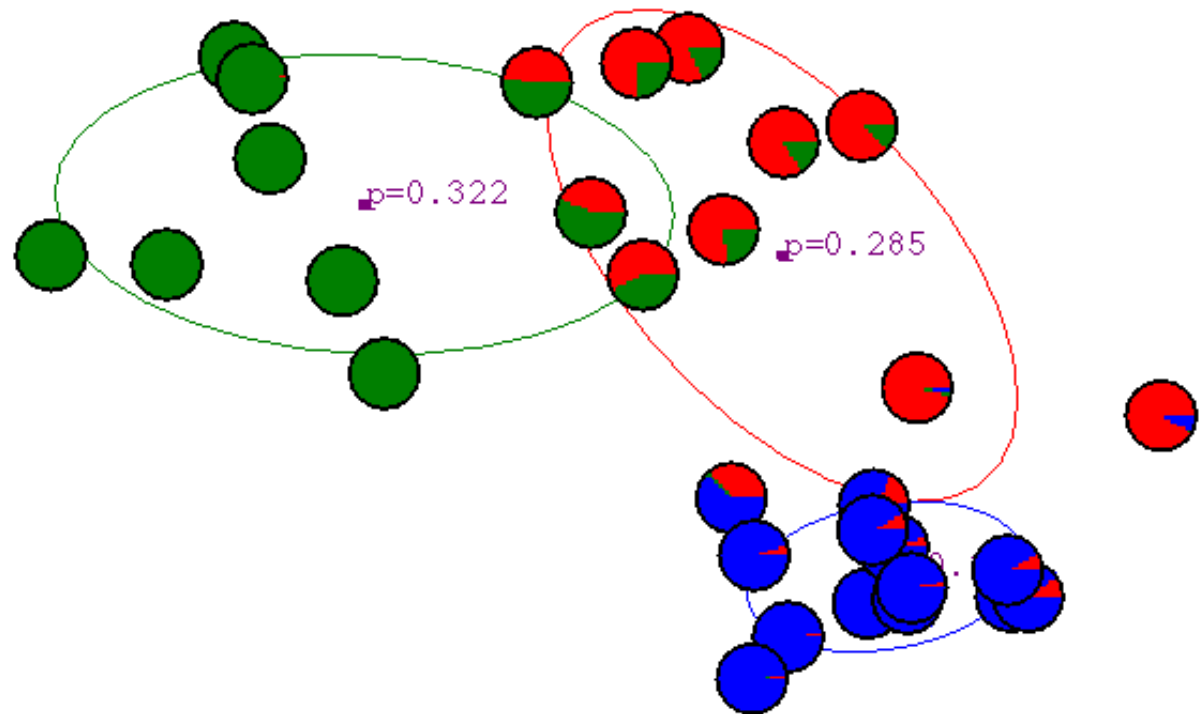


Slide Credit: Andrew W. Moore

After 4th iteration

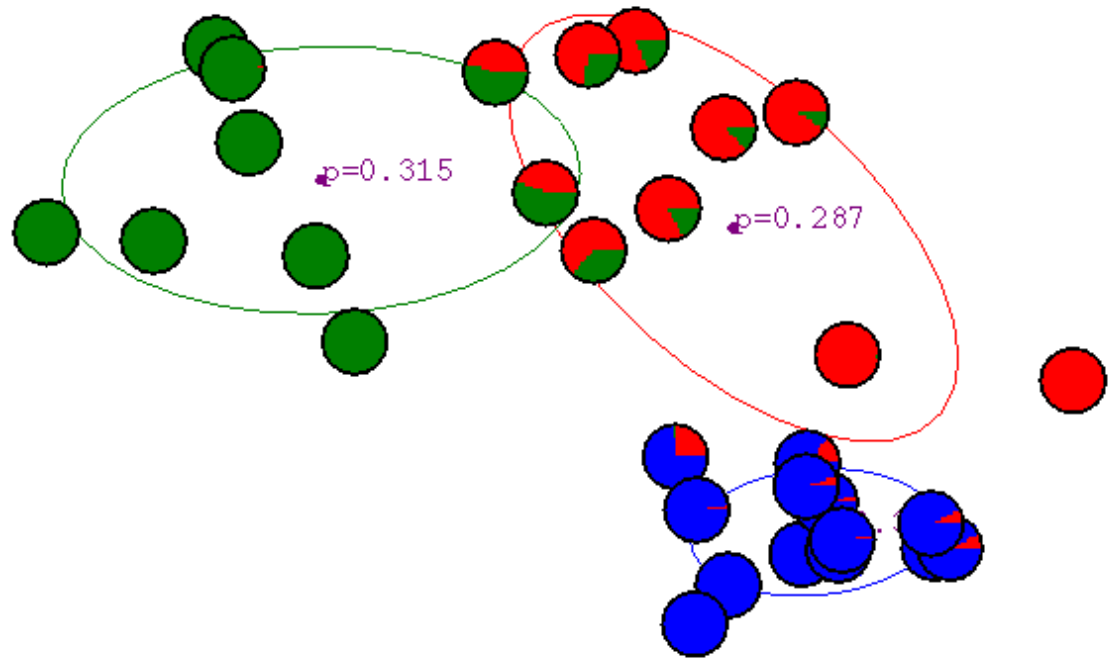


After 5th iteration

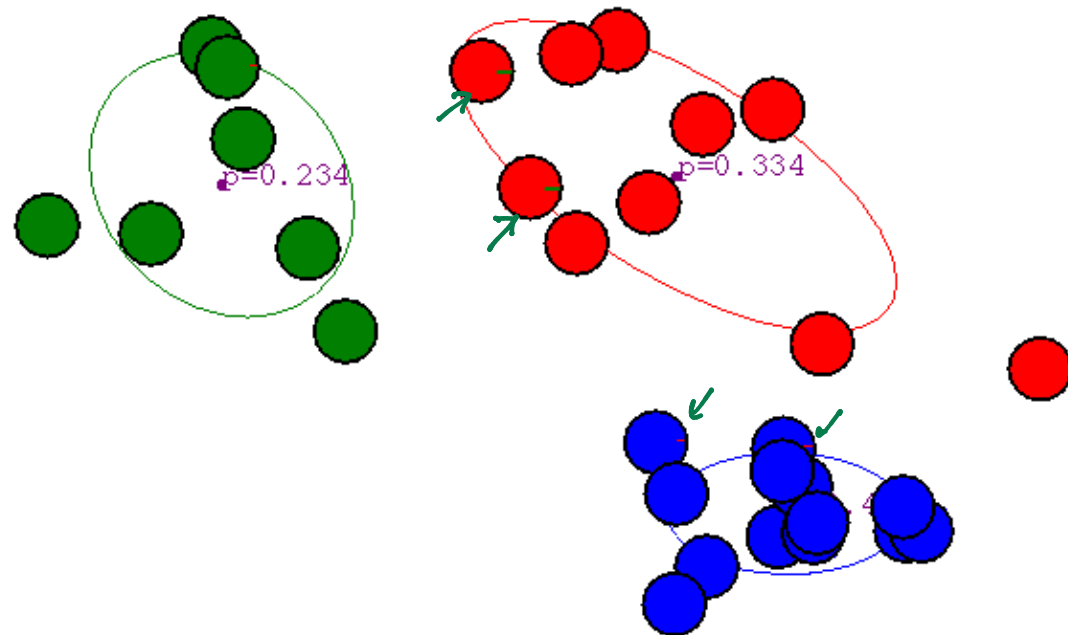


Slide Credit: Andrew W. Moore

After 6th iteration



After 20th iteration



Slide Credit: Andrew W. Moore

$$k=2$$

$$\underline{M_1 \sigma_1}$$

$$k=3$$

$$\underline{M_2 \sigma_2}$$

$$\underline{M_3 \sigma_3}$$

GMM equations

GPM
Sunday



$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z})$$

Mixture models

K = number of mixture components (clusters),

π_j = mixture weights

$p_j(\mathbf{x})$ = family of distributions

(Gaussian, Bernoulli, etc).

$$p(\mathbf{x}) = \sum_{j=1}^K \pi_j p_j(\mathbf{x})$$

$$0 \leq \pi_j \leq 1, \quad \sum_{j=1}^K \pi_j = 1$$

Gaussian Mixture models

$$p(\mathbf{x}|\boldsymbol{\theta}) = \sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$$

$$0 \leq \pi_j \leq 1, \quad \sum_{j=1}^K \pi_j = 1$$

$\boldsymbol{\theta}$ = collection of all the parameters of the model
(mixture weights, means, and covariance matrices):

$$\boldsymbol{\theta} := \{\pi_1, \dots, \pi_K, \boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_K\}$$

General EM algorithm

Observed data: $X = \{x_1, \dots, x_n\}$

Unknown variables: y

In clustering: $y = 1 \dots K$ clusters

Parameters: θ

In GMM : $\theta = \pi : \{\pi_1, \dots, \pi_K\}$

$\mu : \{\mu_1, \dots, \mu_K\}$

$\Sigma : \{\Sigma_1, \dots, \Sigma_K\}$

Goal: $\hat{\theta}_n = \arg \max_{\theta} \log P(X | \theta)$

$$\text{In GMM : } \ln p(\mathbf{X} | \pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

EM algorithm for GMM

E step:

Evaluate the responsibilities (posterior probabilities) using the current parameter

$$\begin{aligned}\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) &= \frac{p(z_k = 1)p(\mathbf{x}|z_k = 1)}{\sum_{j=1}^K p(z_j = 1)p(\mathbf{x}|z_j = 1)} \\ &= \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.\end{aligned}$$

- For each example $\mathbf{x}_n$,
- Compute $\gamma(z_{nk})$ the probability that $\mathbf{x}_n$ is generated by component Z_k i.e it belongs to cluster k
- If $\mathbf{x}_n$ is very likely under the k_{th} Gaussian, it gets high weight

EM algorithm for GMM

M-Step :

- maximize the expectation of the complete-data log-likelihood, computed with respect to the conditional probabilities found in the Expectation step. The result of the maximization is a new parameter vector μ^{new} , Σ^{new} and π^{new}
- Keep $\gamma(z_{nk})$ fixed, and apply MLE for maximizing of $\ln p(\mathbf{X}|\pi, \mu, \Sigma)$ for μ_k , Σ_k and π_k , to get μ^{new} , Σ^{new} and π^{new}

$$\ln p(\mathbf{X}|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

Repeat E & M until convergence

- In practice, the algorithm is deemed to have converged when the change in the log likelihood function, or alternatively in the parameters, falls below some threshold

EM algorithm for GMM

To Estimate:

M-Step

Initialize π, μ, Σ and
also evaluate the log likelihood

E-Step

$$\mu_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n$$

$$\Sigma_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}}) (\mathbf{x}_n - \mu_k^{\text{new}})^T$$

$$\pi_k^{\text{new}} = \frac{N_k}{N}$$

$$N_k = \sum_{n=1}^N \gamma(z_{nk})$$

Perform E-Step Given $\gamma(z_k)$

$$\begin{aligned} \gamma(z_k) &\equiv p(z_k = 1 | \mathbf{x}) = \\ &= \frac{p(z_k = 1) p(\mathbf{x} | z_k = 1)}{\sum_{j=1}^K p(z_j = 1) p(\mathbf{x} | z_j = 1)} \\ &= \frac{\pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \mu_j, \Sigma_j)} \end{aligned}$$

Perform M-Step Given π, μ, Σ

$$p(z_k = 1) = \pi_k$$

Repeat Until
Convergence

N_k = the effective number of points assigned to cluster k

EM Algorithm

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	innovate	achieve	lead
	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

I Standardize the data if required:

II Fix the no.of.cluster expected

III Initialize the prototypes:
Mean, Covariance, Weights

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

V. Calculate the log likelihood of the points

VI Maximization Step: Fix the membership(responsibility matrix) and re-estimate the prototypes

VII Calculate the new log likelihood of the points. Repeat E & M Step till convergence is achieved:

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

Sample calculation for this step is shown below:

$$\begin{aligned}
 \text{Weight1} * N1 &= \\
 0.5 * P(\text{Doc1}, \text{Cluster1}) &= (\text{Doc1}, (3,3), (4,0.707)) \\
 &= 0.5 * \frac{e^{\frac{-(2-3)^2}{2*4^2}}}{4 * \sqrt{2*3.14}} * \frac{e^{\frac{-(2-3)^2}{2*0.707^2}}}{0.707 * \sqrt{2*3.14}} \\
 &= 0.01004
 \end{aligned}$$

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

For Cluster 1 assumed parameters, here for Doc1 apply in the below formula and multiply with the weight = 0.5. For simplicity, assume off diagonal values in covariance matrix as 0 (This is usually done in text analysis use cases). similarly do it for all the Doc's

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

GMM

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Similarly:

For Cluster 2 assumed parameters, here for Doc1 apply in the below formula and multiply with the weight = 0.5. For simplicity, assume off diagonal values in covariance matrix as 0 (This is usually done in text analysis use cases). similarly do it for all the Doc's

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$



	Doc1	Doc2	Doc3

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2



	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Per Docs sum the previous two found values. This gives the denominator of this formula

For Doc2 :

$$\text{Weight1} * \text{N1} = 0.0005$$

$$\text{Weight2} * \text{N2} = 0.0828$$

$$\text{Sum} = 0.08329 = \sim 0.0833$$

$$P(Z_{\text{Doc1-Cluster1}}) = 0.006$$

$$P(Z_{\text{Doc1-Cluster1}}) = 0.994$$

Now find the membership of doc :

P(Zi1) = dividing Weight1*N1 by Sum

P(Zi2) = dividing Weight2*N1 by Sum

Similarly do it for all the Doc's

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2



	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

V. Calculate the log likelihood of the points

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Log likelihood =
 $\ln(0.010) + \ln(0.083) + \ln(0.006)$
 = -12.16

Log likelihood = $\ln(\text{sum of Doc1}) + \ln(\text{sum of Doc2}) + \ln(\text{Sum of Doc3}) \dots$

This value needs to be found for every iteration and at one iteration it will approach a very large value . That will be one of the stopping convergence point of the algorithm.

EM algorithm for GMM

To Estimate:

M-Step

Initialize π, μ, Σ and
also evaluate the log likelihood

E-Step

$$\begin{aligned}\mu_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n \\ \Sigma_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}}) (\mathbf{x}_n - \mu_k^{\text{new}})^T \\ \pi_k^{\text{new}} &= \frac{N_k}{N} \\ N_k &= \sum_{n=1}^N \gamma(z_{nk})\end{aligned}$$

Perform E-Step Given $\gamma(z_k)$

$$\begin{aligned}\gamma(z_k) &\equiv p(z_k = 1 | \mathbf{x}) = \\ &= \frac{p(z_k = 1) p(\mathbf{x} | z_k = 1)}{\sum_{j=1}^K p(z_j = 1) p(\mathbf{x} | z_j = 1)} \\ &= \frac{\pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \mu_j, \Sigma_j)}.\end{aligned}$$

Perform M-Step Given π, μ, Σ

$$p(z_k = 1) = \pi_k$$

Repeat Until
Convergence

N_k = the effective number of points assigned to cluster k

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2



	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

VI Maximization Step: Fix the membership(responsibility matrix) and re-estimate the prototypes

Sample calculations shown below

	Doc1	Doc2	Doc3
P(Zi1)	0.9999	0.006	1
P(Zi2)	0.0001	0.994	0

	Means	SDs	Weight
Cluster 1			
Cluster 2			

$$N_{k=1} = \text{Cluster 1} = \\ = 0.9999 + 0.006 + 1 \\ = 2.0059$$

This is the value N1 represented here

$$\text{MEAN}_{k=1} = \\ (4.49, 2.00) = \sim (4.49, 2)$$

$$\text{Std.Dev}_{k=1} = \\ ([0.9999(2-4.49)^2 + 0.006(4-4.49)^2 + 1(7-4.49)^2]/N1, \\ [0.9999(2-2)^2 + 0.006(5-2)^2 + 1(2-2)^2]/N1) \\ = (6.23, 0.03)$$

$$N_k = \sum_{n=1}^N \gamma(z_{nk})$$

N1 for cluster k=1 is got by summing these. Similarly do it for N2

For cluster k=1 is Means obtained by multiplying the data by the membership.

$$[0.9999(2) + 0.006(4) + 1(7)]/N1, [0.9999(2) + 0.006(5) + 1(2)]/N1$$

$$\mu_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n$$

Similarly do it for cluster 2

$$\Sigma_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}}) (\mathbf{x}_n - \mu_k^{\text{new}})^T$$

$$\pi_k^{\text{new}} = \frac{N_k}{N}$$

$$\text{Weight}_{k=\text{cluster1 (new)}} \\ = 2.0059/3 = 0.67$$

GMM - EM Algorithm

2nd Iteration continues



	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

I Standardize the data if required:

II Fix the no.of.cluster expected

III Initialize the prototypes:
Mean, Covariance, Weights

IV Expectation-Step: Fix prototype
& find the membership of each
point weighted by the probability
value

V. Calculate the log likelihood of the
points

VI Maximization Step: Fix the
membership(responsibility matrix)
and re-estimate the prototypes

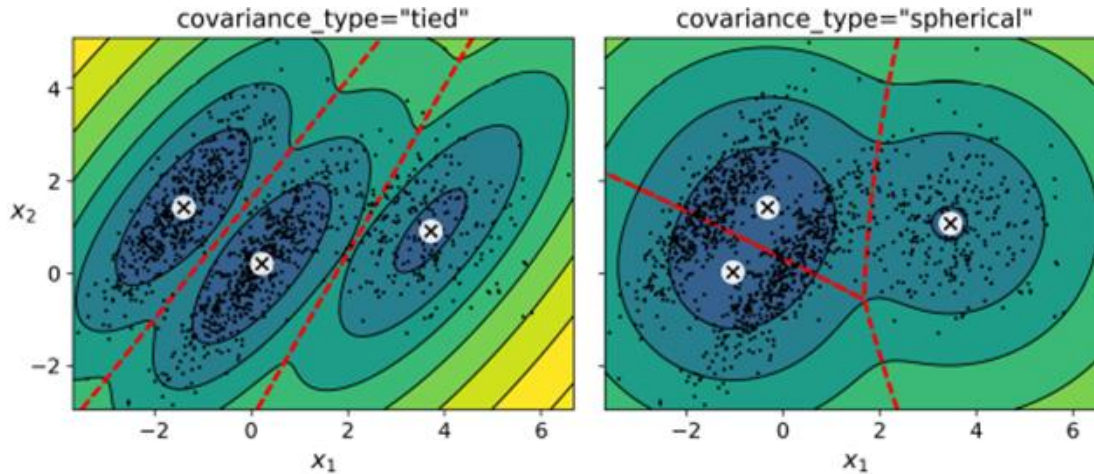
VII Calculate the new log likelihood
of the points. Repeat E & M Step till
convergence is achieved:

$$\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Means	SDs	Weight
Cluster 1	(4.49,2)	(6.32,0.03)	0.67
Cluster 2	(3.99,4.99)	(0.001,0.001)	0.33

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Gaussian Mixture Model



Note:

The GMM model can be relaxed by setting these parameters while calling the function

- **"spherical"**: all clusters must be spherical, but they can have different diameters (i.e., different variances).
- **"diag"**: clusters can take on any ellipsoidal shape of any size, but the ellipsoid's axes must be parallel to the coordinate axes (i.e., the covariance matrices must be diagonal).
- **"tied"**: all clusters must have the same ellipsoidal shape, size and orientation (i.e., all clusters share the same covariance matrix).

Gaussian Mixture Model

```
from sklearn.mixture import GaussianMixture
```

```
gm = GaussianMixture(n_components=3, n_init=10)  
gm.fit(X)
```

```
gm.weights_  
gm.means_  
gm.covariances_
```

```
>>> gm.predict(X)  
array([2, 2, 1, ..., 0, 0, 0])  
>>> gm.predict_proba(X)  
array([[2.32389467e-02, 6.77397850e-07, 9.76760376e-01],  
       [1.64685609e-02, 6.75361303e-04, 9.82856078e-01],
```

References

- Christopher Bishop: Pattern Recognition and Machine Learning, Springer International Edition - Chapter 9
- <https://www.youtube.com/watch?v=TG6Bh-NFhA0>
- <https://www.youtube.com/watch?v=qMTuMa86NzU>